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Numerical simulation of 2-D seismic wave propagation in the presence of a topographic fluid–solid interface at the sea bottom by the curvilinear grid finite-difference method

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SUMMARY

This study simulates seismic wave propagation across a 2-D topographic fluid (acoustic) and solid (elastic) interface at the sea bottom by the finite-difference method (FDM). In this method, seismic waves in sea water are governed by acoustic wave equations, whereas seismic waves in solid earth are governed by elastic wave equations. The fluid–solid interface condition is implemented on the interface. Body-conforming grids are used to fit the topographic fluid–solid interface which naturally avoids spurious diffractions due to staircase approximation. A collocated-grid MacCormack FDM is utilized to update the wavefields in the fluid and solid media. The fluid–solid interface condition is explicitly implemented by decomposing the velocity and stress components to the normal and tangential directions with respect to the interface within a fourth-order Runge–Kutta time-marching scheme. The algorithm solutions for both flat and topographic fluid–solid interface models are compared with analytical solutions and spectral element solutions to validate the proposed method. Results show a suitable agreement with the reference solutions and hence confirms the validity of this method. The proposed FDM enforces the numerical solutions to satisfy the exact interface condition and it is more accurate than the conventional FDM that uses effective media parameters to approximate the interface condition.

Key words: Interface waves; Wave propagation; Wave scattering and diffraction.

1 INTRODUCTION

The fluid–solid problem in this study is about seismic waves that travel through the interface between inviscid fluid and solid media. The fluid–solid interface condition shows that the traction and normal component of particle velocity across the interface are continuous. By contrast, the tangential components of the particle velocity may be discontinuous. A chief benefits of correctly implementing the fluid–solid interface is the opportunity to investigate the physical phenomena associated with the interface. These phenomena include the generation and propagation of T-waves (e.g. de Groot-Hedlin & Orcutt 2001; Okal 2008; Jamet *et al.* 2013), Scholte and leaky Rayleigh waves (e.g. de Hoop & van der Hijden 1983, 1984; Padilla *et al.* 1999; Zhu & Popovics 2004; Carcione & Helle 2004; Zheng *et al.* 2013) as well as seismic scattering caused by the seafloor (e.g. Robertsson & Levander 1995; Greaves & Stephen 2000). A thorough understanding of the Scholte wave can help seismic exploration studies infer the reliable shear wave velocity below the seafloor (e.g. Bohlen *et al.* 2004; Kugler *et al.* 2007; Muyzert 2007a,b). Another benefit is the ability of the approach to help in analysing specific seismic data close to the ocean (e.g. Noguchi *et al.* 2007) or data from multicomponent seismic ocean acquisition. The acquired information can then be used to obtain enhanced reservoir images for characterization (Farfour & Yoon 2016).

Analytical (or semi-analytical) and numerical methods have been applied to solve the fluid–solid problem. Analytical (or semi-analytical) methods, such as the generalized reflection/transmission coefficient method (GRTM; e.g. Luco & Apsel 1983; Chen 1990), can calculate synthetic waveforms for layered or quasi-layered fluid–solid media (Chen 1990, 1995; Bouchon *et al.* 1995; Chen 1996; Ge & Chen 2007; Qian & Yamanaka 2012) with high accuracy and efficiency. However, handing complex media remains a significant challenge for these methods. By contrast, numerical methods can treat complex media more easily than the above-mentioned methods. However, the numerical methods cannot readily obtain accurate solutions for the fluid–solid interface problem. Hence many numerical methods have been further investigated to solve the problem that arises in scientific research and engineering studies. These numerical methods include the finite-difference method (FDM; e.g. Bayliss *et al.* 1986; Graves 1996; Moczo *et al.* 2002; Robbert *et al.* 2002; Lombard & Piraux 2004; Okamoto & Takenaka 2005), pseudospectral method (e.g. Tessmer *et al.* 1992; Hung & Forsyth 1998; Feng *et al.* 2007), finite-element method (FEM; e.g. Bermúdez *et al.* 1999; Zhang 2004), discontinuous Galerkin

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method (e.g. Käser & Dumbser 2008; Wilcox *et al.* 2010), finite-volume method (e.g. Voinovich *et al.* 2003) and spectral element method (SEM; e.g. Komatitsch *et al.* 2000; Chaljub *et al.* 2003).

Among these numerical methods, the FDM is a more direct approximation for numerical solutions of wave equations than the methods based on other formulations. The FDM also demands a lower computing cost and fits parallel computing. However, the accurate calculation of wavefield propagation solutions that involve interfaces, such as free surface topography and fluid–solid interface, is a challenge for finite difference. Previous studies on FDMs that involve fluid–solid and free surface interfaces may be catalogued into different classes in terms of grids (i.e. Cartesian grid and curvilinear grid). For Cartesian grid cases, the fluid–solid interfaces are discretized by stair-case approximation which may cause numerical scattering (Muir *et al.* 1992). The interface conditions are implemented by effective media (e.g. Robertsson 1996; Moczo *et al.* 2007), which assume that the tangential component of particle velocity is continuous across the interface and conflicts with the fluid–solid interface conditions. Another approach based on the Cartesian grid is the immersed-interface method (also called embedded-boundary method). This method can handle free-surface (e.g. Kreiss & Petersson 2006; Lombard *et al.* 2008) and fluid–solid interfaces (Lombard & Piraux 2004) in a way that allows high-order accuracy as conjectured by the approach's proponents. However, a recent study on free surface interface based on the aforementioned method shows that instability may arise in a 3-D scenario(Gao *et al.* 2015). For curvilinear grid cases, the grid lines align with the topographic interface. This occurrence not only avoids spurious diffraction naturally (e.g. Zhang & Chen 2006; Appelö & Petersson 2009; Tarrass *et al.* 2011; Puente *et al.* 2014), but renders implementing boundary conditions convenient. Recent studies on curvilinear grids may be distinguished from one another by the algorithms for implementing boundary conditions and grid schemes. These algorithms include the collocated scheme (also called non-staggered or unstaggered schemes) (e.g. Zhang & Chen 2006; Appelö & Petersson 2009; Tarrass *et al.* 2011; Zhang *et al.* 2012b), partly staggered scheme (e.g. Ely *et al.* 2008), and fully staggered scheme (e.g. Puente *et al.* 2014). Zhang & Chen (2006) utilized the collocated scheme and traction image method to implement the free surface boundary condition in the first-order wave equation. Appelö & Petersson (2009) also proposed a stable method on the collocated scheme to implement the free surface boundary condition in the second-order wave equation. Puente *et al.* (2014) employed a fully staggered Lebedev scheme and a mimetic approach to solve the free surface condition in the first-order wave equation. The curvilinear grids and collocated scheme are utilized to solve the fluid–solid interface problem for algorithm proposed in the following sections.

The current study extends the curvilinear grid finite difference (Zhang & Chen 2006) to simulate seismic wave propagation in 2-D fluid–solid media with a topographic interface. The curvilinear grids are adopted to conform to the topographic fluid–solid interface. Seismic waves in sea water are governed by first-order acoustic wave equations, whereas those in solid earth are by first-order velocity-stress elastic wave equations. The fluid–solid interface condition is implemented explicitly by decomposing the velocity and stress components to normal and tangential directions on the interface. The computing results of this algorithm for flat and topographic fluid–solid interface models are compared by analytical solutions and spectral element solutions. The results show a suitable agreement with the reference solutions, which conforms the validity of the proposed method. The proposed FDM enforces the numerical solutions to satisfy the exact condition and it is more accurate than the conventional FDM that uses effective media parameters (e.g. Graves 1996; Robbert *et al.* 2002; Moczo *et al.* 2007) to implement the interface condition; it is also demonstrated in the flat fluid–solid interface model section. Finally, a true sea-bottom model with heterogeneities in the media parameters is built to show wavefield complexity. Its synthetic waveforms are also compared with the SEM solutions. When the fluid–solid interface is complex, the proposed curvilinear grid FDM (Curv. FDM) would employ finer grids to discretize the interface to maintain the accuracy; this scenario makes the Curv. FDM require more computing resources than those of the Cartesian grid FDM. Nevertheless, the discontinuous curvilinear grid techniques proposed by Li *et al.* (2015) can be applied to this problem so that the proposed method can largely save on computational resources by utilizing fine grids for the interface area and coarse grids for other areas.

2 BASIC EQUATIONS

For convenience of description, the Einstein summation convention for repeated indices is adopted and the derivatives with respect to the *y*-axis (into the paper) are assumed to be zero.

2.1 Body-conforming grid and coordinate transformation

When the topographic fluid–solid interface is given, the curvilinear grids are adopted to discretize the geological model such that the interface grid line coincides with the topographic fluid–solid interface (Fig. 1, left panel). This grid is called the boundary-conforming (or body-fitting) grid, which has been widely applied in seismic wave modelling (e.g. Fornberg 1988; Tessmer *et al.* 1992; Hestholm & Ruud 1994, 1998; Zhang & Chen 2006; Appelö & Petersson 2009; Zhang *et al.* 2012b; Sun *et al.* 2016). The grid qualities, such as orthogonality of grid points, concentration of grid points and smoothness of the variations in the grid size, can be controlled during the generations (Thompson *et al.* 1985) by algebraic or partial differential equation methods. The body-conforming grid can be generated by software (e.g. ICEM CDF software of ANSYS Inc. and Gridgen) with specified grid qualities. The proposed algorithm can improve its computing efficiency with several special grid qualities (e.g. orthogonality of grid points). However, the algorithm can still work properly without such additions. For simplicity, the body-confirming grids in this study are generated by simple shrinking or stretching the grid lines in vertical direction. For finite difference computing, the generated grids in the physical space (x, z)

$$x = x(\xi, \zeta) \tag{1a}$$

$$z = z(\xi, \zeta) \tag{1b}$$

should first be mapped to a uniform rectangular grid (Fig. 1) in the computational space (ξ, ζ) by the coordinate transformation (Thompson *et al.* 1985). Then, the coefficients $\partial_\xi x$, $\partial_\zeta x$, $\partial_\xi z$ and $\partial_\zeta z$ for the Jacobian of the coordinate transformation

$$J = \partial_\xi x \partial_\zeta z - \partial_\zeta x \partial_\xi z, \tag{2}$$

should be numerically calculated. By contrast, the coefficients $\partial_x \xi$, $\partial_z \xi$, $\partial_x \zeta$ and $\partial_z \zeta$ are calculated by the chain rules. Details on the coefficient calculation are the same as those reported by Zhang & Chen (2006).

2.2 Seismic wave equations in fluid and solid media

The fluid in this study is assumed to be inviscid and governed by the following isotropic acoustic wave equation (e.g. Landau &

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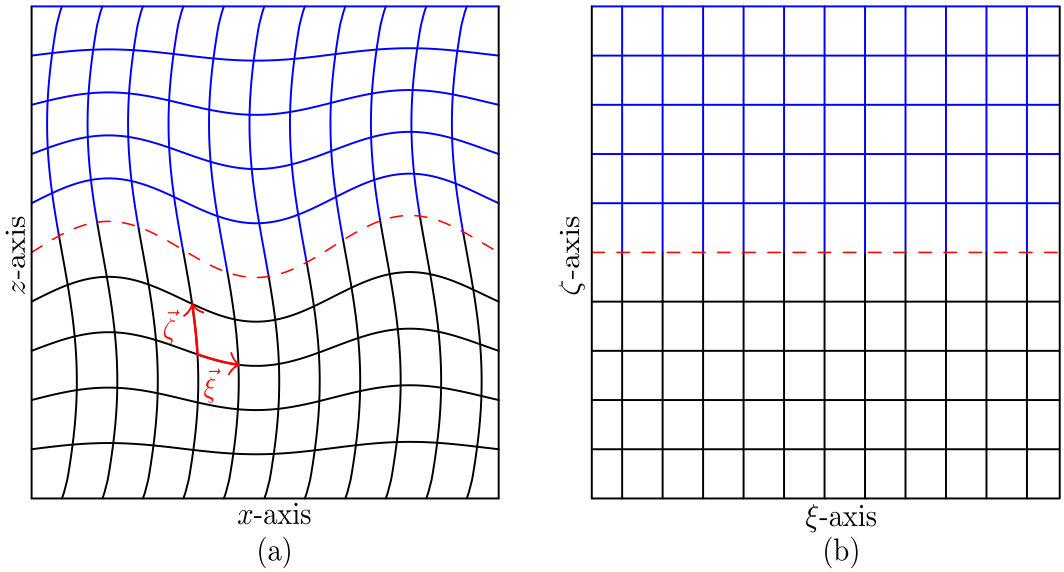


Figure 1. Geometric illustration of the body-conforming grids (a) in the physical space and rectangle grids and (b) in the computational space. The dashed line in (a) coincides with the fluid–solid interface. The lines above and below the dashed lines in (a) and (b) represent the grids in the fluid and solid media, respectively. The physical grids are transformed into the computational grids by the coordinate transformation.

Lifshitz 1959)

$$\rho_f \partial_t v_i^f + \partial_i p = 0 \tag{3a}$$

$$\partial_t p + \kappa \partial_i v_i^f = 0 \tag{3b}$$

where index $i \in \{x, z\}$, ρ_f is the mass density of fluid media, κ is the bulk modulus of the fluid media, v_i^f is the particle velocity vector component of the fluid media and p is the hydrostatic pressure disturbance. The solid is assumed to be linear isotropic elastic and the elastic wave equation may be stated in the following form:

$$\rho_s \partial_t v_i^s = \partial_j \sigma_{ij} \tag{4a}$$

$$\partial_t \sigma_{ij} = \lambda \partial_k v_k^s \delta_{ij} + \mu(\partial_j v_i^s + \partial_i v_j^s) \tag{4b}$$

where the indices $i, j, k \in \{x, z\}$, ρ_s is the mass density of the solid media, λ and μ are the Lamé constants, v_i^s is the particle velocity vector component of the solid media and σ_{ij} is the stress tensor component.

The 2-D wave equations of the fluid and solid media in the curvilinear coordinate (ξ, ζ) can be derived from eqs (3) and (4) by the chain rules. Their full forms are presented as follows:

$$\rho_f \partial_t v_x^f + \partial_x \xi \partial_\xi p + \partial_x \zeta \partial_\zeta p = 0 \tag{5a}$$

$$\rho_f \partial_t v_z^f + \partial_z \xi \partial_\xi p + \partial_z \zeta \partial_\zeta p = 0 \tag{5b}$$

$$\partial_t p + \kappa(\partial_x \xi \partial_\xi v_x + \partial_x \zeta \partial_\zeta v_x + \partial_z \xi \partial_\xi v_z + \partial_z \zeta \partial_\zeta v_z) = 0 \tag{5c}$$

and

$$\rho_s \partial_t v_x^s = \partial_x \xi \partial_\xi \sigma_{xx} + \partial_x \zeta \partial_\zeta \sigma_{xx} + \partial_z \xi \partial_\xi \sigma_{xz} + \partial_z \zeta \partial_\zeta \sigma_{xz} \tag{6a}$$

$$\rho_s \partial_t v_z^s = \partial_x \xi \partial_\xi \sigma_{xz} + \partial_x \zeta \partial_\zeta \sigma_{xz} + \partial_z \xi \partial_\xi \sigma_{zz} + \partial_z \zeta \partial_\zeta \sigma_{zz} \tag{6b}$$

$$\partial_t \sigma_{xx} = \chi(\partial_x \xi \partial_\xi v_x + \partial_x \zeta \partial_\zeta v_x) + \lambda(\partial_z \xi \partial_\xi v_z + \partial_z \zeta \partial_\zeta v_z) \tag{6c}$$

$$\partial_t \sigma_{zz} = \lambda(\partial_x \xi \partial_\xi v_x + \partial_x \zeta \partial_\zeta v_x) + \chi(\partial_z \xi \partial_\xi v_z + \partial_z \zeta \partial_\zeta v_z) \tag{6d}$$

$$\partial_t \sigma_{xz} = \mu(\partial_x \xi \partial_\xi v_z + \partial_x \zeta \partial_\zeta v_z) + \mu(\partial_z \xi \partial_\xi v_x + \partial_z \zeta \partial_\zeta v_x) \tag{6e}$$

where $\chi = \lambda + 2\mu$.

Eqs (5) and (6) can also be expressed in the following matrix form:

$$\partial_t \boldsymbol{W} = \boldsymbol{A} \partial_\xi \boldsymbol{W} + \boldsymbol{B} \partial_\zeta \boldsymbol{W} \tag{7}$$

where $\boldsymbol{W}$ is the velocity-stress vector and $\boldsymbol{A}$, $\boldsymbol{B}$ are coefficient matrices (see Appendix A for details).

The momentum equations in eqs (5) and (6) can be written in the conservative form of the ∇ operator (Thompson *et al.* 1985; Zhang & Chen 2006) as follows:

$$J\rho_f \partial_t v_x^f + \partial_\xi(J \partial_x \xi p) + \partial_\zeta(J|\nabla\zeta| T_x^f) = 0 \tag{8a}$$

$$J\rho_f \partial_t v_z^f + \partial_\xi(J \partial_z \xi p) + \partial_\zeta(J|\nabla\zeta| T_z^f) = 0 \tag{8b}$$

and

$$J\rho_s \partial_t v_x^s = \partial_\xi(J \partial_x \xi \sigma_{xx} + J \partial_z \xi \sigma_{xz}) + \partial_\zeta(J|\nabla\zeta| T_x^s) \tag{9a}$$

$$J\rho_s \partial_t v_z^s = \partial_\xi(J \partial_x \xi \sigma_{xz} + J \partial_z \xi \sigma_{zz}) + \partial_\zeta(J|\nabla\zeta| T_z^s) \tag{9b}$$

where the tractions in the fluid and solid media are as follows:

$$\begin{aligned} \mathbf{T}^f &= \left(T_x^f, T_z^f\right) = (\partial_x \zeta p/|\nabla\zeta|, \partial_z \zeta p/|\nabla\zeta|) \\ \mathbf{T}^s &= \left(T_x^s, T_z^s\right) = ((\partial_x \zeta \sigma_{xx} + \partial_z \zeta \sigma_{xz})/|\nabla\zeta|, \\ &\qquad (\partial_x \zeta \sigma_{xz} + \partial_z \zeta \sigma_{zz})/|\nabla\zeta|) \end{aligned} \tag{10}$$

and $|\nabla\zeta| = \sqrt{(\partial_x \zeta)^2 + (\partial_z \zeta)^2}$.

2.3 Fluid–solid interface condition

The fluid–solid interface condition that couples the acoustic wave equation (eq. 3) and elastic wave equation (eq. 4) is described as a dynamic boundary condition

$$\sigma_{ij} n_i = -p I_{ij} n_i \tag{11}$$

and a kinematic continuity condition

$$v_i^s n_i = v_i^f n_i \tag{12}$$

where, indices $i, j \in \{x, z\}$, n_i denotes the component of the normal vector of the fluid–solid interface and I_{ij} is the unit second-order

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tensor. The dynamic boundary condition ensures that the traction across the fluid–solid interface is continuous and the kinematic continuity condition describes that only the normal component of the particle velocity across the interface is continuous. This condition implies that the fluid contact with the solid can only slide but not be pulled apart (Chapman 2004).

Given the 2-D assumption, the dynamic boundary and kinematic continuity conditions are written in the 2-D curvilinear coordinate as follows:

$$\sigma_{nn} = -p \tag{13a}$$

$$\sigma_{n\tau} = 0 \tag{13b}$$

and

$$v_n^s = v_n^f \tag{14}$$

where σ_{nn}, $\sigma_{n\tau}$, v_n^s and v_n^f are the wavefield components in the local coordinate (see Appendix B for details).

3 CURVILINEAR GRID FINITE-DIFFERENCE SCHEME FOR THE PROBLEM

The curvilinear grid finite-difference scheme is a type of collocated grid MacCormack-type finite-difference operators with fourth-order Runge–Kutta time-marching scheme. This scheme has been applied in seismic wave modelling (Zhang & Chen 2006; Zhang *et al.* 2012a,b) and earthquake dynamics (Zhang *et al.* 2014b). A brief introduction to the curvilinear grid finite-difference scheme is presented in this section (see Zhang & Chen (2006) and Zhang *et al.* (2012a,b) for details).

3.1 MacCormack finite-difference operators

The collocated and staggered grids are two types of grid schemes, which are widely implemented in finite difference. The collocated grid is preferable over the staggered grid for boundary-conforming problem because this grid can avoid the complex interpolations of the staggered grid (Magnier *et al.* 1994; Igel *et al.* 1995). As such, the collocated grid can hold the overall accuracy. The MacCormack-type collocated-grid finite-difference scheme (Bayliss *et al.* 1986; Xie & Yao 1988; Vdfidus *et al.* 1992; Dai *et al.* 1995) is widely utilized in seismic modelling. Its wide usage is due to the inherent dissipation (artificial damping) of one-sided differences it introduces to eliminate the instability of collocated-grid central difference caused by odd-even decoupling. The MacCormack-type scheme alternately utilizes forward and backward one-sided differences at the stages of a multistage Runge–Kutta time integration. As a result, the central difference is recovered when the one-sided differences are added together at the last time stage. The DRP/opt MacCormack, 4-2 MacCormack and 2-2 MacCormack operators (Hixon 1997) are used in the algorithm for spatial derivatives in this study. The forward and backward one-sided difference operators of DRP/opt MacCormack finite difference can be expressed as follows (i.e. ξ-axis):

$$L_\xi^F(\boldsymbol{W})_i = \frac{1}{\Delta\xi}\sum_{n=-1}^{3} a_n \boldsymbol{W}_{i+n} \tag{15a}$$

$$L_\xi^B(\boldsymbol{W})_i = \frac{1}{\Delta\xi}\sum_{n=-1}^{3} -a_n \boldsymbol{W}_{i-n} \tag{15b}$$

where L_ξ represents the spatial difference with respect to the ξ-axis; the subscript i is the grid index; the superscript F and B denote the forward and backward operators, respectively; and $a_{-1} = -0.30874$, $a_0 = -0.6326$, $a_1 = 1.2330$, $a_2 = -0.3334$ and $a_3 = 0.04168$ (Hixon 1997). The forward and backward finite one-sided difference operators of the 4-2 MacCormack scheme can be expressed as follows (i.e. ξ-axis):

$$L_\xi^F(\boldsymbol{W})_i = \frac{1}{\Delta\xi}\sum_{n=0}^{2} a_n \boldsymbol{W}_{i+n} \tag{16a}$$

$$L_\xi^B(\boldsymbol{W})_i = \frac{1}{\Delta\xi}\sum_{n=0}^{2} -a_n \boldsymbol{W}_{i-n} \tag{16b}$$

where the coefficients $a_0 = -7/6$, $a_1 = 8/6$ and $a_2 = -1/6$ (Hixon 1997). The forward and backward one-sided difference operators of the 2-2 MacCormack scheme can be expressed as follows (i.e. ξ-axis):

$$L_\xi^F(\boldsymbol{W})_i = \frac{1}{\Delta\xi}\sum_{n=0}^{1} a_n \boldsymbol{W}_{i+n} \tag{17a}$$

$$L_\xi^B(\boldsymbol{W})_i = \frac{1}{\Delta\xi}\sum_{n=0}^{1} -a_n \boldsymbol{W}_{i-n} \tag{17b}$$

where $a_0 = -1.0$ and $a_1 = 1.0$ (Hixon 1997).

The right-hand side of eq. (7) can be discretized by the following equation:

$$\hat{L}^{FF}(\boldsymbol{W}) = \boldsymbol{A} L_\xi^F(\boldsymbol{W}) + \boldsymbol{B} L_\zeta^F(\boldsymbol{W}) \tag{18a}$$

$$\hat{L}^{BB}(\boldsymbol{W}) = \boldsymbol{A} L_\xi^B(\boldsymbol{W}) + \boldsymbol{B} L_\zeta^B(\boldsymbol{W}) \tag{18b}$$

where F or B in the first and second superscripts over $\hat{L}$ indicates the forward or backward difference, respectively, with respect to the ξ- and ζ-axes, respectively.

3.2 Runge–Kutta time-marching scheme

The fourth-order Runge–Kutta scheme is utilized to update the wavefields from time step n to $n+1$ by the following equation:

$$h^{(1)} = \Delta t \hat{L}^{FF}(\boldsymbol{W}^n) \quad \boldsymbol{V}^{(1)} = \boldsymbol{W}^n + \alpha_2 h^{(1)} \tag{19a}$$

$$h^{(2)} = \Delta t \hat{L}^{BB}(\boldsymbol{V}^{(1)}) \quad \boldsymbol{V}^{(2)} = \boldsymbol{W}^n + \alpha_3 h^{(2)} \tag{19b}$$

$$h^{(3)} = \Delta t \hat{L}^{FF}(\boldsymbol{V}^{(2)}) \quad \boldsymbol{V}^{(3)} = \boldsymbol{W}^n + \alpha_4 h^{(3)} \tag{19c}$$

$$h^{(4)} = \Delta t \hat{L}^{BB}(\boldsymbol{V}^{(3)}) \tag{19d}$$

$$\boldsymbol{W}^{n+1} = \boldsymbol{W}^n + \beta_1 h^{(1)} + \beta_2 h^{(2)} + \beta_3 h^{(3)} + \beta_4 h^{(4)} \tag{19e}$$

where $\boldsymbol{W}^n$ and $\boldsymbol{W}^{n+1}$ are the wavefields at time step n and $n+1$, respectively, and $\alpha_2 = 0.5$, $\alpha_3 = 0.5$, $\alpha_4 = 1.0$, $\beta_1 = 1/6$, $\beta_2 = 1/3$, $\beta_3 = 1/3$ and $\beta_4 = 1/6$. The aforementioned fourth-order Runge–Kutta scheme can be expressed as follows:

$$\boldsymbol{W}^{n+1} = \hat{L}^{FF}\hat{L}^{BB}\hat{L}^{FF}\hat{L}^{BB}\boldsymbol{W}^n. \tag{20}$$

The forward and backward one-sided differences are alternately applied at the stages of the Runge–Kutta integration, and the central difference is obtained at the last step of the scheme. The following two-step time marching sequence

$$\boldsymbol{W}^{n+1} = \hat{L}^{FF}\hat{L}^{BB}\hat{L}^{FF}\hat{L}^{BB}\boldsymbol{W}^n \tag{21a}$$

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$$\boldsymbol{W}^{n+2} = \hat{L}^{BB}\hat{L}^{FF}\hat{L}^{BB}\hat{L}^{FF}\boldsymbol{W}^{n+1} \tag{21b}$$

is employed in this study to avoid numerical bias.

3.3 Gaussian filter

Velocity wavefields may be discontinuous across the fluid–solid interface. Thus, the spatial velocity derivatives with respect to the interface for grid points that are exactly on the interface are always calculated by a one-sided (forward or backward MacCormack 2-2) difference operator to avoid utilizing values across the discontinuous interface. The velocity derivatives calculated in this manner are reasonable. However, the numerical experiments show that this effect may cause instability in the interface after a long computing time. A Gaussian filter (Kristek *et al.* 2010; Zhang *et al.* 2013) is then applied on the interface grid points along the ξ-axis to stabilize the computing for a long time. The filter can be expressed as follows:

$$w_k = A \cdot \frac{\alpha}{\sqrt{\pi}2n} e^{-(\frac{\alpha k}{2n})^2} \tag{22}$$

where $\alpha = 4.2$, the grid spacing ratios are $n = 1$ and A is a constant for normalizing w_k as follows:

$$\sum_{k=-2n}^{2n} w_k = 1. \tag{23}$$

The filter works on the ξ-axis direction at each time step and for a wavefield variable Φ with the grid index i_0 on the interface grids points. Its filtered value $\tilde{\Phi}$ is obtained as follows:

$$\tilde{\Phi}(i_0) = \sum_{i=-2}^{2} w_{i+i_0}\Phi(i+i_0). \tag{24}$$

The numerical experiments in Section 4 show that the filter can stabilize the computation and negligibly affects the physical solutions.

3.4 Implementation of the fluid–solid interface condition

The curvilinear grid line aligns with the fluid–solid interface, which is assumed to be parallel to the ξ-axis and labelled by the grid index k_0 in Fig. 2. The wavefield components and media parameters for the acoustic wave equation are defined on the grid points with $k \geq k_0$. By contrast, those for the elastic wave equation are defined on the grid points with $k \leq k_0$. The grid points on the fluid–solid interface possess the grid index with $k = k_0$ and contains the wavefield components and media parameters for the acoustic and elastic wave equations. A grid point on the fluid–solid interface is split into fluid and solid points to explicitly implement the fluid–solid interface condition (Fig. 2). The split fluid point contains the wavefield components and media parameters only for the fluid media. On the contrary, the split solid point harbours those for the solid media. Notably, the split fluid and solid points for a grid point on the fluid–solid interface share the same spatial location. However, those points infinitely approximate the interface from the fluid and

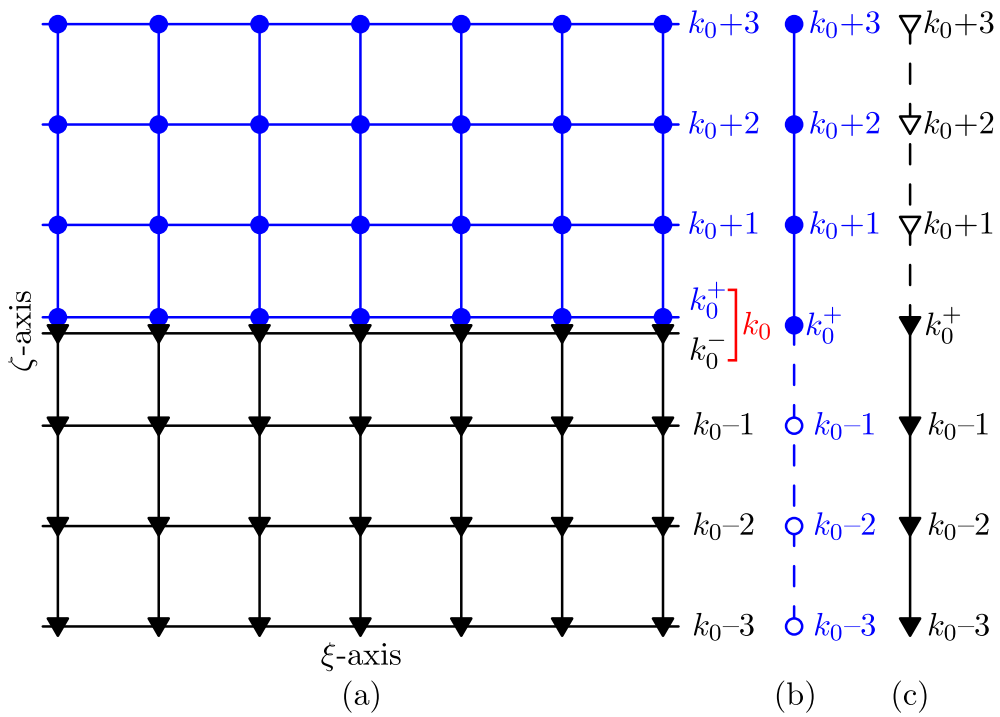


Figure 2. Collocated grids and ghost points near the fluid–solid interface in the computational domain. (a) The collocated grids in the computational domain. The filled circle points represent the grid points that define the wavefields and media parameters for the fluid media, whereas the filled inverted triangle points represent the grid points that define the wavefield components and media parameters for the solid media. The grid points along the ξ-axis with the grid index k_0 represent the fluid–solid interface. The grid points are then split into the fluid points with grid index k_0^+ and solid points with grid index k_0^-. The split fluid and solid points for a grid point on the interface share the same spatial locations. However, these points infinitely approximate the interface from the fluid and solid sides, respectively. (b) Example of ghost points for the fluid media. The empty circles with the grid indices $k_0 - 1$, $k_0 - 2$ and $k_0 - 3$ represent the ghost points for the fluid media. (c) Example of the ghost points for the solid media. The empty inverted triangles with the grid indices $k_0 + 1$, $k_0 + 2$ and $k_0 + 3$ represent the ghost points for the solid media. Tractions are defined on the ghosts points. Notably, the tractions defined on the ghost points are not physically real and are only for the computational purposes.

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Table 1. Misfit quantification of the curvilinear grid FDM for flat fluid–solid interface with PPW = 8.57.

Name	x(m)	z(m)	EM(v_x) (per cent)	PM(v_x) (per cent)	EG(v_x)	PG(v_x)	EM(v_z) (per cent)	PM(v_z) (per cent)	EG(v_z)	PG(v_z)
Rec. 1	600	0	0	0	10	10	4.77	1.20	9.53	9.88
Rec. 2	600	−40	4.00	0.55	9.61	9.94	3.06	0.77	9.69	9.92
Rec. 3	600	−80	5.45	0.42	9.46	9.95	4.40	0.91	9.56	9.91
Rec. 4	600	−120	4.79	0.44	9.53	9.95	5.19	1.03	9.49	9.89
Rec. 5	600	−160	4.38	0.53	9.57	9.94	5.11	1.12	9.50	9.88
Rec. 6+	600	−200	4.66	0.54	9.54	9.94	2.02	0.34	9.80	9.97
Rec. 6−	600	−200	4.05	0.73	9.60	9.92	4.77	0.79	9.53	9.90
Rec. 7	600	−240	3.79	0.73	9.63	9.92	4.66	0.72	9.54	9.92
Rec. 8+	700	−200	7.68	1.74	9.25	9.82	4.47	0.66	9.56	9.93
Rec. 8−	700	−200	5.48	0.67	9.46	9.93	5.61	0.82	9.45	9.91
Rec. 9+	800	−200	6.43	1.82	9.37	9.81	3.36	0.45	9.67	9.96
Rec. 9−	800	−200	3.95	0.81	9.61	9.92	5.95	0.79	9.42	9.92

'+', the receiver is on the fluid side of the interface.
'−', the receiver is on the solid side of the interface.

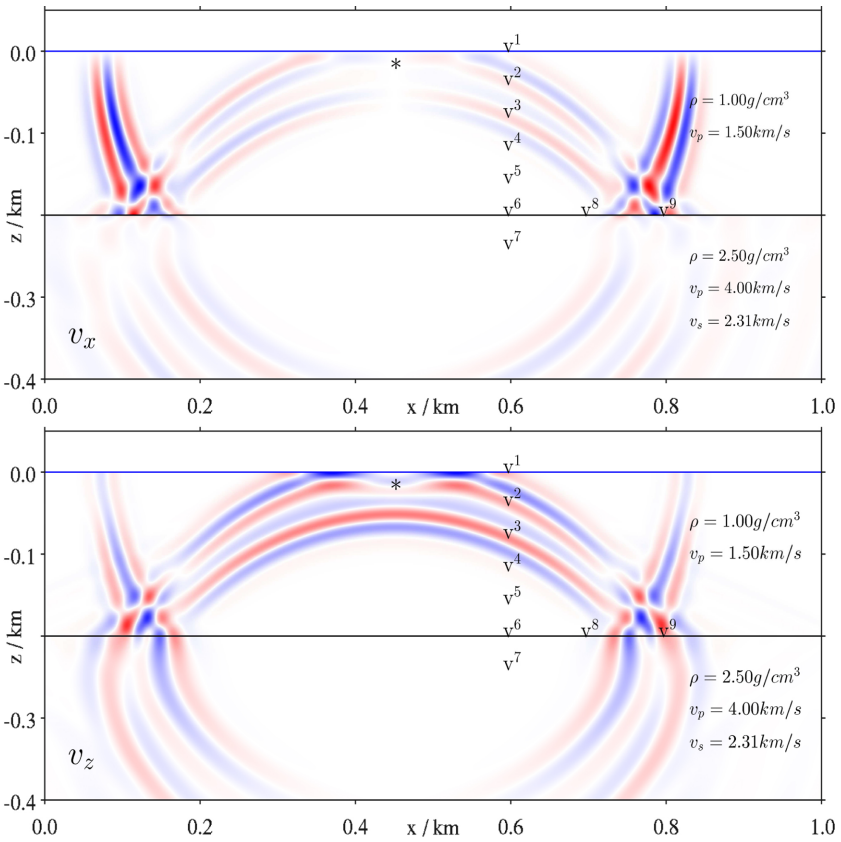


Figure 3. Snapshots of the v_x and v_z components at $t = 0.3$ s for the flat fluid–solid interface model. The black line along the x-axis in the middle of each figure ($z = -0.2$ km) represents the flat fluid–solid interface, whereas the line along the x-axis at the top of each figure ($z = 0.0$ km) represents the flat free surface. The explosive source is marked by a black star in the fluid media, and the receivers are marked by the v-symbols with specified numbers. The media parameters for each of the layers are shown on the right side of the snapshots.

solid sides, respectively. For convenience, the grid points on the interface are called interface points and labelled by the grid index k_0. Meanwhile, the split fluid and solid points along the ξ-axis are labelled as k_0^+ and k_0^-, respectively, to avoid confusion. The grid points, which are three or more points away the interface points in the ζ-axis direction, are called interior grid points. By contrast, those that are one or two points away from the interface points in the ζ-axis direction are called near interface points. Special treatments should be applied to handle the interface points and their neighbouring points. The wavefield components in the fluid and solid media are updated by eq. (7), except that the velocity components on the interface and near interface grid points are updated by eqs (8) and (9).

The spatial derivatives with respect to the ξ-axis on all grid points and those with respect to the ζ-axis on the interior grid points are calculated by the DRP/opt MacCormack finite difference. The spatial derivatives of the velocity components with respect to the ζ-axis on the grid points with two points away from the interface points are calculated by the 4-2 MacCormack finite difference. By contrast, those one point away from the interface points are calculated by the 2-2 MacCormack finite difference. The derivatives of the velocity components with respect to the ζ-axis on the

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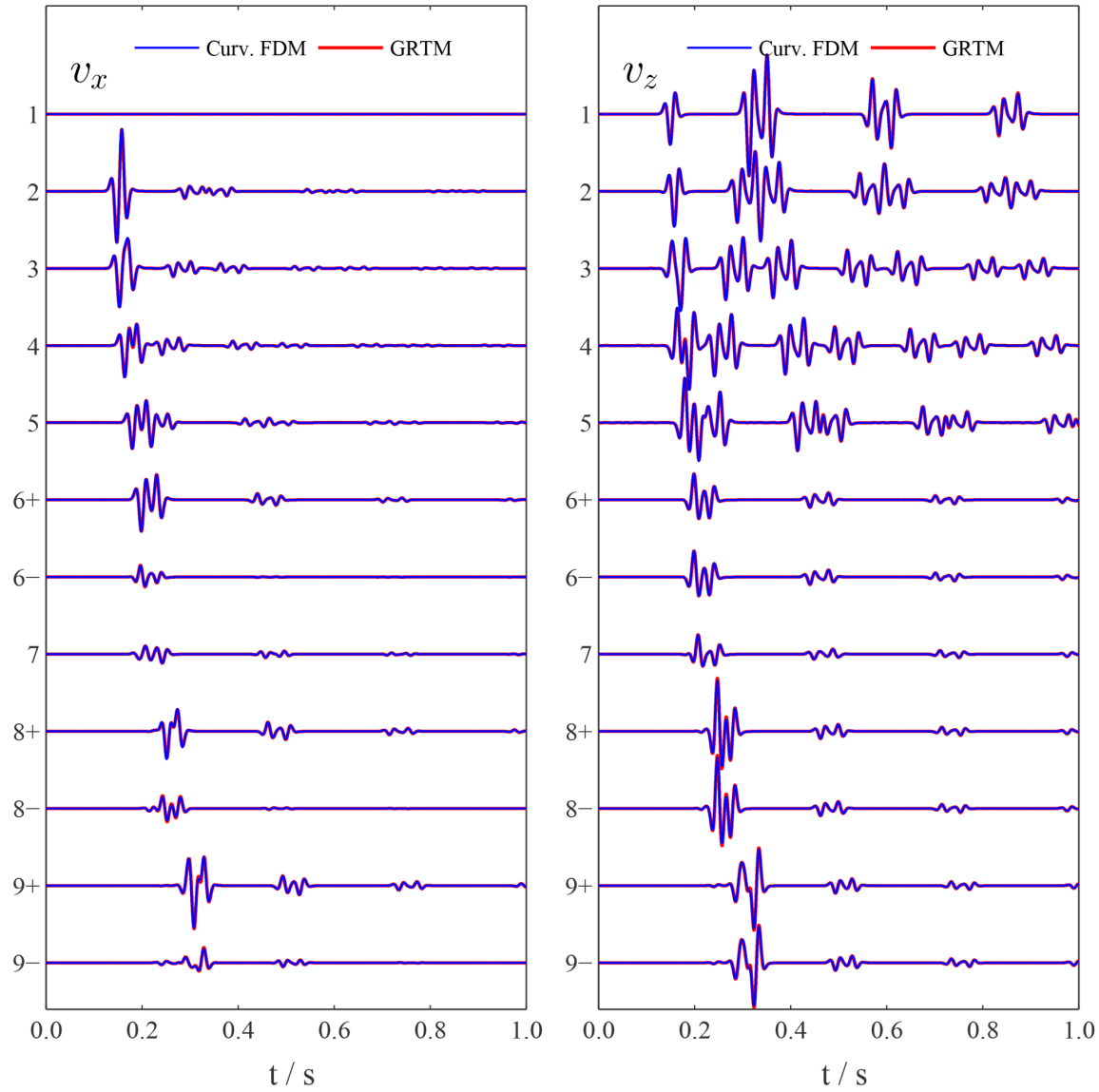


Figure 4. Synthetic seismograms for the receivers in Fig. 3. The left part shows the v_x components, whereas the right part shows the v_z components. The thick lines are computed by the GRTM, whereas the thin lines are computed by the curvilinear grid FDM. Receivers 6, 8 and 9 are located on the fluid–solid interface, and the symbols '+' and '−' represent the fluid side and solid side of the interface, respectively.

interface grid points are calculated by the forward and backward 2-2 MacCormack finite differences and fluid–solid interface conditions.

Given that the fluid–solid interface condition does not ensure that the spatial derivatives of tractions across the interface are continuous, we calculate their one side (fluid or solid side) spatial derivatives with respect to the ζ-axis near the fluid–solid interface from the same side information. Therefore, we utilize the following central symmetric extrapolated formula for tractions in the ghost points as follows:

$$\mathbf{T}^f|_{k_0^+-n} = 2\mathbf{T}^f|_{k_0^+} - \mathbf{T}^f|_{k_0^++n}, \quad n = 1, 2, 3 \tag{25}$$

for the fluid side and

$$\mathbf{T}^s|_{k_0^-+n} = 2\mathbf{T}^s|_{k_0^-} - \mathbf{T}^s|_{k_0^--n}, \quad n = 1, 2, 3 \tag{26}$$

for the solid side. The extrapolated formula is inspired by the traction image method proposed by Zhang & Chen (2006), but the values of the central points in this formula may change with time. The spatial derivatives of the traction components with respect to the ζ-axis in eqs (8) and (9) on the interface and near interface grid points are calculated by the DRP/opt MacCormack finite difference.

The fluid–solid interface condition is implemented on the interface points within the fourth-order Runge–Kutta integration through the following processors. (1) $h^{(1)}$ is calculated in eq. (19) to obtain the components of $h^{(1)}$ in the local coordinate by rotation transformation in Appendix B. (2) If $h^{(1)}$ is calculated by the L^{FF} operator, then $\partial_t v_n^s$ and $\partial_t \sigma_{nn}$ are replaced by $\partial_t v_n^f$ and $-\partial_t p$j, and $\partial_t \sigma_{n\tau} = 0$ is set to ensure the interface condition. (3) If $h^{(1)}$ is calculated by the L^{BB} operator, then $\partial_t v_n^f$ and $\partial_t p$ are replaced by $\partial_t v_n^s$ and $-\partial_t \sigma_{nn}$, and $\partial_t \sigma_{n\tau} = 0$ is set to ensure the interface condition. (4) The modified components of $h^{(1)}$ are rotated into the Cartesian coordinate to obtain $h^{(1)*}$. (5) The Runge–Kutta processor is implemented to perform the same operations on $h^{(2)}$, $h^{(3)}$ and $h^{(4)}$ and to obtain $h^{(2)*}$, $h^{(3)*}$ and $h^{(4)*}$. (6) $\mathbf{W}^{n+1*} = \mathbf{W}^n + \beta_1 h^{(1)*} + \beta_2 h^{(2)*} + \beta_3 h^{(3)*} + \beta_4 h^{(4)*}$ is calculated, and the components of $\mathbf{W}^{n+1*}$ are filtered by the Gaussian filter in eq. (24). (7) The filtered components of $\mathbf{W}^{n+1*}$ are rotated into the local coordinate, and the values of v_n^f, v_n^s

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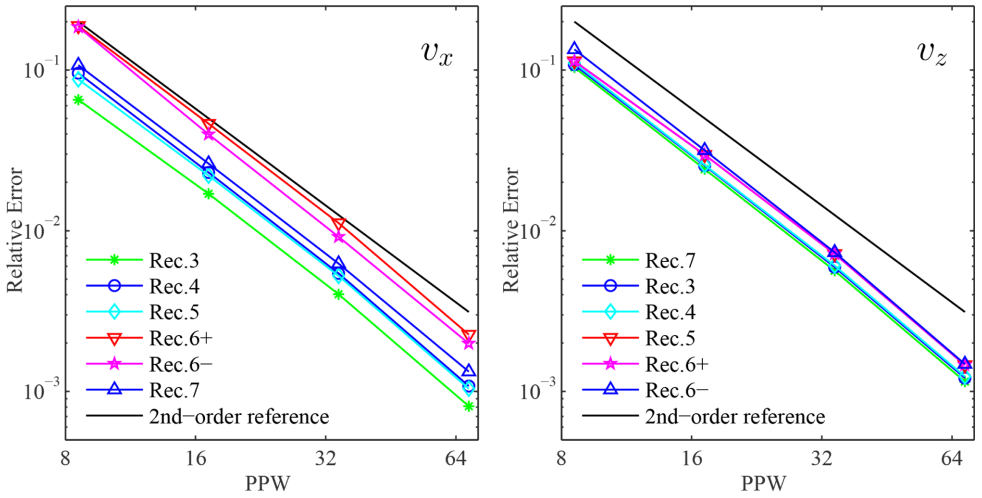


Figure 5. Relative errors of receivers 3 to 7 in Fig. 3 with different PPWs for the flat fluid–solid interface model. The left part shows the relative errors of the v_x components, whereas the right part shows those of the v_z components. The black line is a reference on whose slope is valued at 2. The slope of each line is approximately 2, which shows that the spatial accuracy of the algorithm is a second order.

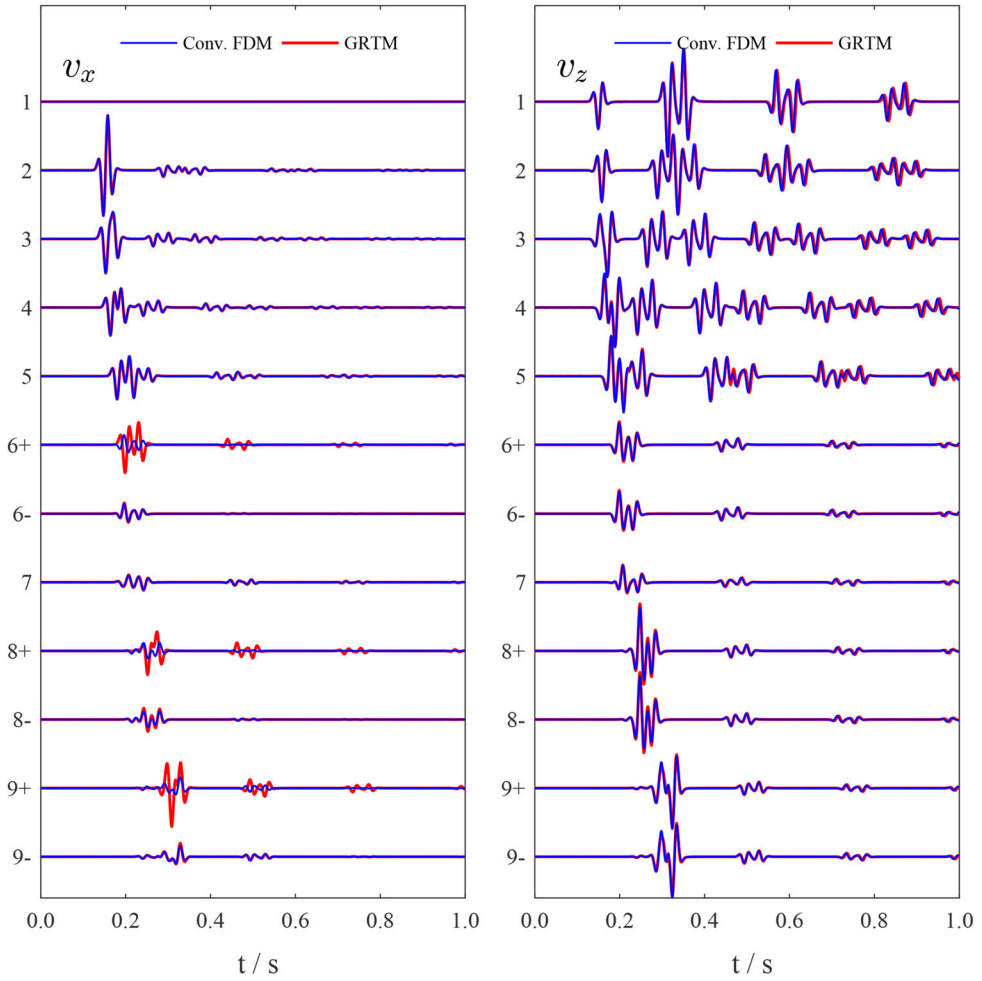


Figure 6. Synthetic seismograms for the receivers in Fig. 3. The left part shows the v_x components, whereas the right part shows the v_z components. The thick lines are computed by GRTM, whereas the thin lines are computed by the conventional FDM (Conv. FDM) that utilizes effective media parameters to implement the fluid–solid interface condition.

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Table 2. EM and PM of the conventional FDM solutions with different PPWs in percentage.

Error type	PPW	Rec. 5	Rec. 6+	Rec. 6−	Rec. 7	Rec. 8+	Rec. 8−	Rec. 9+	Rec. 9−
EM (v_x)	∼8	5.14	70.0	6.20	7.83	75.6	12.3	76.0	10.7
	∼16	2.99	71.3	5.09	3.22	73.3	5.91	74.3	4.91
	∼32	0.78	70.6	2.79	1.34	72.1	3.01	73.3	2.25
PM (v_x)	∼8	3.33	85.0	2.99	6.90	47.8	3.11	43.0	2.55
	∼16	1.48	86.4	1.22	2.57	44.5	1.26	41.2	1.16
	∼32	0.71	86.3	0.94	1.44	43.5	0.79	40.5	0.75
EM (v_z)	∼8	11.6	3.94	9.31	5.60	4.23	7.36	5.68	10.3
	∼16	6.22	1.17	2.83	2.07	2.43	4.21	3.58	6.50
	∼32	2.24	0.30	0.77	0.78	1.22	2.08	1.64	2.95
PM (v_z)	∼8	12.6	3.15	7.53	7.00	2.40	4.21	2.28	4.08
	∼16	5.73	1.33	3.26	3.21	1.38	2.41	1.34	2.44
	∼32	2.82	0.64	1.65	1.59	0.76	1.33	0.75	1.37

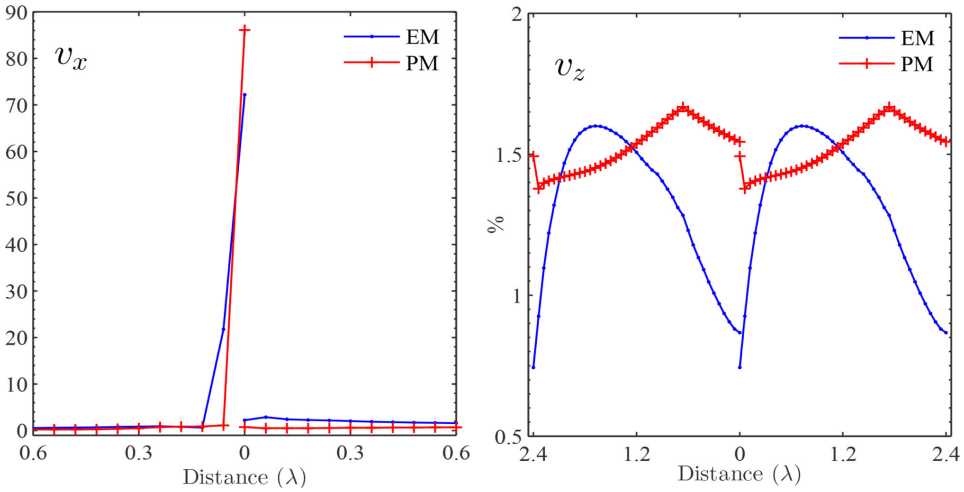


Figure 7. EM and PM errors of the conventional finite-difference solutions with PPW = 32 for the receivers on the grid points between receivers 5 and 7 in Fig. 3. The horizontal axis shows the distance in wavelength ($\lambda = 17.1$ m) between the receiver and flat interface. The points on the left side of 0 point in the horizontal axis represent the receivers in the fluid media, whereas those on the right side of 0 points represent the receivers in the solid media.

Table 3. Misfit quantification of the curvilinear grid FDM for canyon-shaped fluid–solid interface with PPW = 8.82.

Name	x(m)	z(m)	EM(v_x) (per cent)	PM(v_x) (per cent)	EG(v_x)	PG(v_x)	EM(v_z) (per cent)	PM(v_z) (per cent)	EG(v_z)	PG(v_z)
Rec. 1	400	0.0	0	0	10	10	4.58	2.25	9.55	9.77
Rec. 2	400	−42.6	2.42	0.44	9.76	9.95	2.47	0.50	9.75	9.94
Rec. 3	400	−79.4	2.68	0.45	9.73	9.95	1.84	0.46	9.81	9.95
Rec. 4	400	−122.3	1.83	0.37	9.81	9.96	3.00	0.90	9.70	9.90
Rec. 5	400	−159.1	2.18	0.32	9.78	9.96	2.90	0.69	9.71	9.93
Rec. 6	400	−201.9	1.16	0.44	9.88	9.95	2.97	0.71	9.70	9.92
Rec. 7	400	−238.7	1.83	0.83	9.82	9.91	4.09	0.71	9.59	9.92
Rec. 8	400	−281.6	3.38	0.92	9.67	9.90	3.96	1.88	9.61	9.81
Rec. 9	400	−318.4	2.27	0.48	9.77	9.95	6.60	1.08	9.36	9.89
Rec. 10	400	−361.3	3.26	0.57	9.67	9.94	4.27	0.90	9.58	9.91

and σ^f_{nn}, $-p$ are replaced by $(v^f_n + v^s_n)/2$ and $(\sigma_{nn} - p)/2$ to ensure the interface conditions. The modified components of $\mathbf{W}^{n+1*}$ are rotated to the Cartesian coordinate to obtain the updated $\mathbf{W}^{n+1}$.

The lower-order finite-difference operators (eqs 16 and 17) are utilized to calculate the spatial derivatives of the particle velocity components with respect to the ζ-axis near or on the fluid–solid interface. These operators notably lower the spatial accuracy to the second order.

4 NUMERICAL EXPERIMENTS

We compare the results of the flat fluid–solid interface model with the GRTM solutions (Chen 1999) and those of the irregular fluid–solid interface model with SEM solutions (Komatitsch *et al.* 2000) to verify the algorithm. The comparisons are quantitatively assessed by the global measurement of single-valued envelope misfit (EM), phase misfit (PM), envelope goodness-of-fit (EG) and phase goodness-of-fit (PG) of the time-frequency (TF) misfit criteria proposed by Kristeková *et al.* (2006, 2009). EM and PM show the disagreements of the two solutions in terms of the amplitude and phase, respectively. By contrast, EG and PG show the agreements of the two solutions in terms of the amplitude and phase, respectively. The TF misfit criteria can properly distinguish between the amplitude and phase errors. The criteria have been widely employed in seismology to evaluate numerical solutions (e.g. Pérez-Ruiz *et al.* 2007; Fichtner & Igel 2008; Käser *et al.* 2008;

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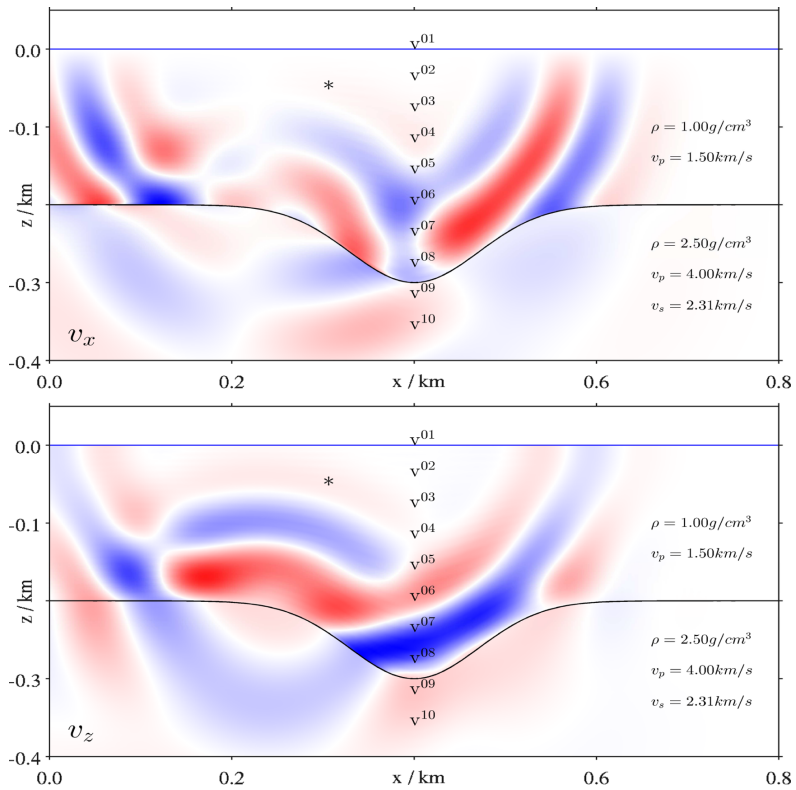


Figure 8. Snapshots of the v_x and v_z components at $t = 0.3$ s for the canyon-shaped fluid–solid interface model. The interface is approximated by $z = z_0 + b_0 \exp[-(x - x_0)^2/a_0^2]$, where $a_0 = 100$ m, $b_0 = 100.0$ m, $x_0 = 2000.0$ m and $z_0 = -200.0$ m. The black curve in the middle of each figure represents the topographic fluid–solid interface. The line along the x-axis at the top of each figure ($z = 0.0$ km) represents the flat free surface. The explosive source is marked by a black star in the fluid media, and the receivers are marked by v-symbols with specified numbers. The media parameters for each of the layers are shown on the right side of the snapshots.

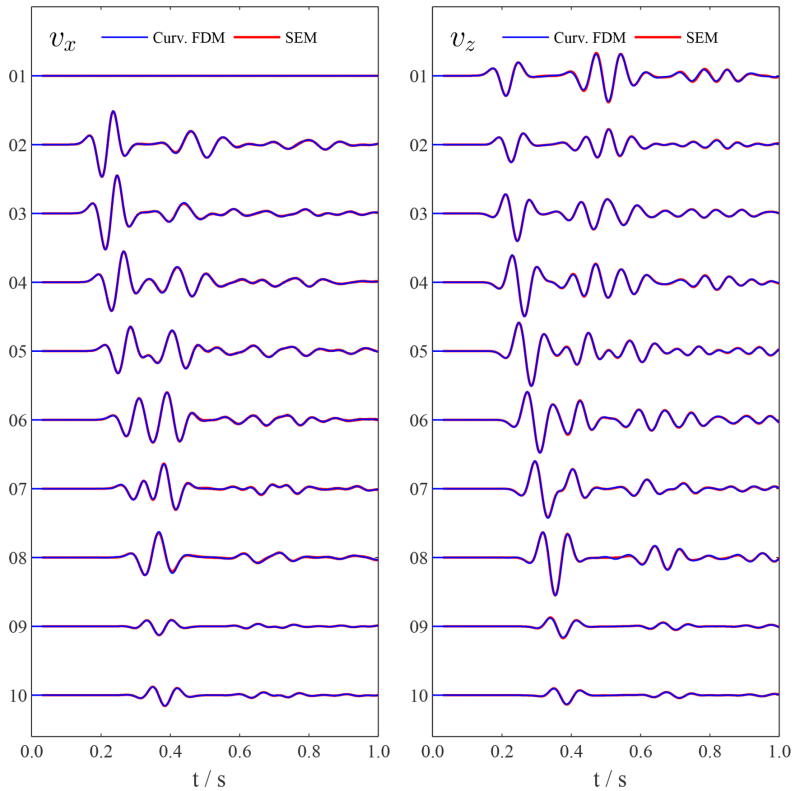


Figure 9. Synthetic waveforms for the receivers in Fig. 8. The left part shows the v_x components, whereas the right part shows the v_z components. The thick lines are computed by SEM, and the thin lines are computed by the curvilinear grid FDM.

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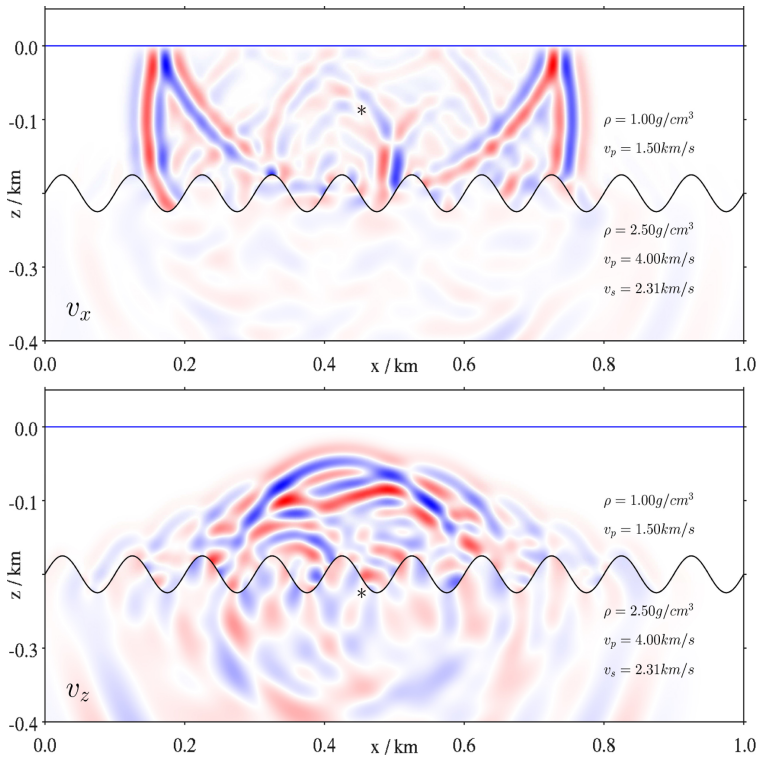


Figure 10. Snapshots of the v_x and v_z components for the sinusoidal fluid–solid interface. The black curve in the middle of each snapshot represents the fluid–solid interface, which is approximated by $z = z_0 + a_0 \sin[2\pi(x - x_0)/b_0]$, where $z_0 = -200.0$ m, $x_0 = 500.0$ m, $a_0 = 25.0$ m and $b_0 = 100.0$ m. The line at the top of each snapshot ($z = 0.0$ km) represents the flat free surface. The black star that represents the location of the source and media parameters are shown in the snapshots. The v_x snapshot is taken at 0.25 s with an explosive source located in the fluid media. By contrast, the v_z snapshot is taken at 0.15 s with an explosive source located in the solid media.

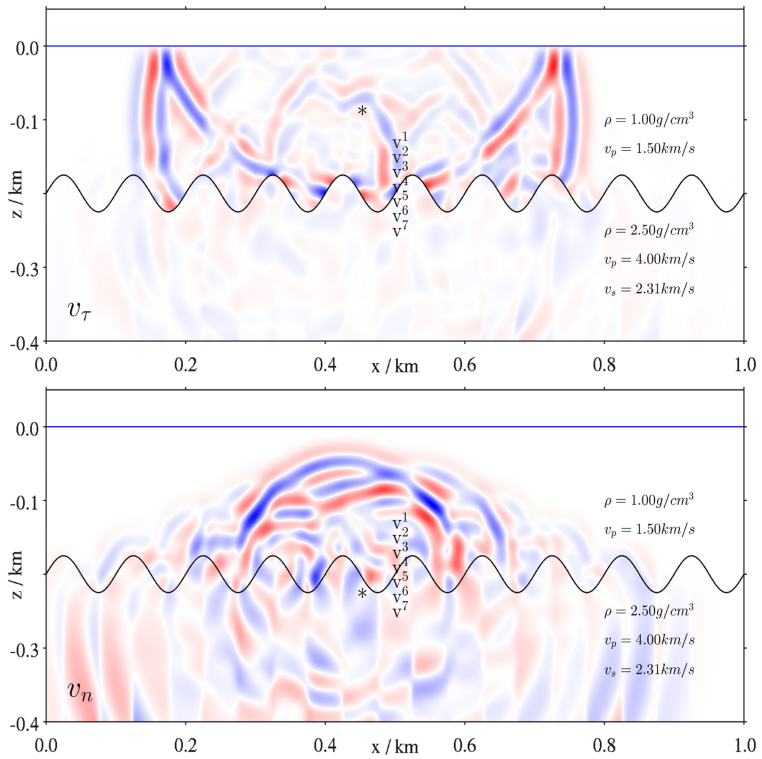


Figure 11. Snapshots of the v_τ and v_n components, which are rotated from their v_x and v_z components in Fig. 10 by formulas in Appendix B. The black star represents the source location, and the v-symbols with specified numbers represent the location of the receivers. The media parameters are written in the snapshots. Receiver 4 is located on the fluid–solid interface. The v_τ snapshot is taken at 0.16 s with an explosive source located in the solid media. Meanwhile, the v_n snapshot is taken at 0.30 s with an explosive source located in the fluid media.

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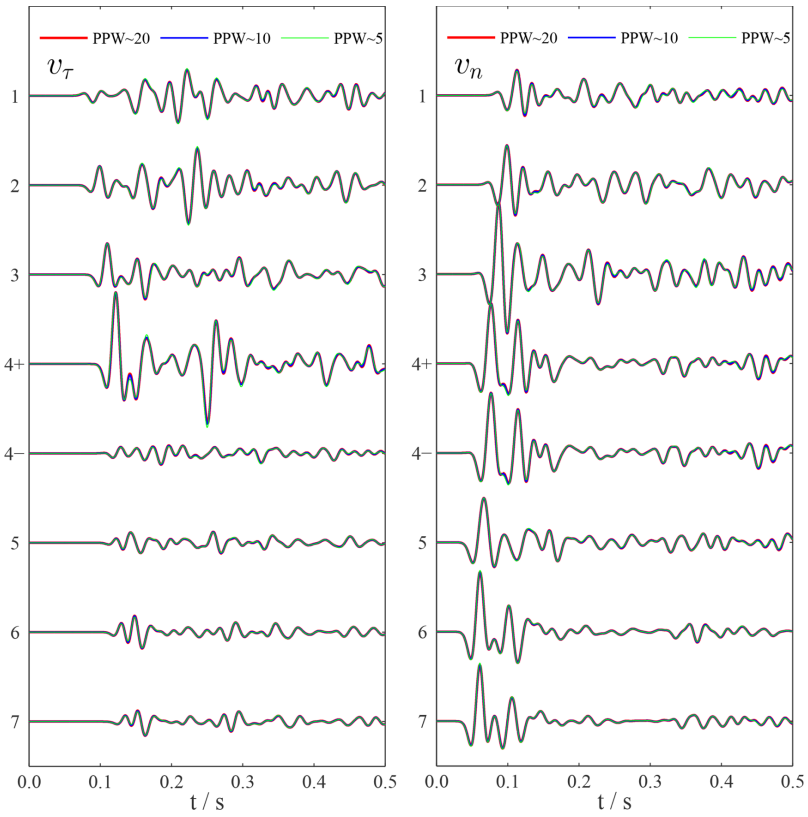


Figure 12. Synthetic v_τ and v_n waveforms for the receivers in Fig. 11 with different PPWs. The different line styles represent the different PPWs. Quantitative comparisons are listed in Table 4.

Table 4. Misfit quantification of the curvilinear grid FDM for sinusoidal-shaped fluid–solid interface.

Error type	PPW	Rec. 1	Rec. 2	Rec. 3	Rec. 4+	Rec. 4−	Rec. 5	Rec. 6	Rec. 7
EM(v_τ)	5	7.21	7.02	3.45	5.81	7.28	2.61	3.39	3.22
	10	1.64	1.59	0.79	1.29	1.64	0.57	0.75	0.70
PM(v_τ)	5	1.22	1.29	0.89	1.75	1.91	0.48	0.61	0.61
	10	0.29	0.29	0.19	0.39	0.55	0.09	0.14	0.12
EM(v_n)	5	7.52	6.67	5.64	6.48	2.77	6.26	5.38	5.36
	10	1.71	1.52	1.27	1.46	0.63	1.44	1.22	1.13
PM(v_n)	5	0.64	0.67	1.01	1.18	0.51	1.14	1.23	1.67
	10	0.16	0.17	0.23	0.25	0.11	0.24	0.27	0.37

Zhang *et al.* 2012b). The maximum value of the 'excellent' level given by Kristeková *et al.* (2009) for EM is 0.22, whereas that for PM is 0.2. EM or PM should be under their maximum 'excellent' levels. The minimum value of the 'excellent' level given by Kristeková *et al.* (2009) for EG or PG is 8, and these factors should be larger than 8.

The Ricker wavelet is selected in the following experiments as the source time function. It can be expressed as $s(t) = (1 - 2\pi^2 f_c^2(t - t_0)^2)\exp(-\pi^2 f_c^2(t - t_0)^2)$, where f_c is the centre frequency and t_0 is the delay time. The points per wavelength (PPW) is computed by $v_s/(2.5 f_c h_s)$, where v_s is the minimum velocity and h_s is the maximum spatial interval value of the adjacent grid nodes in the x- and z-axes directions. The free surface of the fluid media is flat and the free surface boundary conditions are implemented by the traction image method (Zhang & Chen 2006). The outgoing waves can be absorbed by ADE CFS-PML (Berenger 1994; Chew & Liu 1996; Wang & Liang 2006; Zhang & Yang 2010; Zhang *et al.* 2014a) or the absorbing boundary layers (Cerjan *et al.* 1985). For simplicity, 10 absorbing boundary layers are built to absorb the outgoing waves. The grids for the curvilinear finite difference is generated by stretching or shrinking the grid lines in the z-axis direction. Given that fluid–solid wavefields across its interface may be discontinuous, the waveforms of the receivers on the interface are labelled with '+' and '−' to represent the fluid and solid sides of the interface, respectively.

4.1 Half-space model with the flat fluid–solid interface

The first test is a half-space with the flat fluid–solid interface at $z = -200.0$ m in the physical domain. An explosive source is located at (x: 450 m, z: −30 m) in the fluid media with $f_c = 35$ Hz and $t_0 = 0.05$ s. The spatial locations of the receivers are shown in Table 1. The source-receiver geometry and media parameters are displayed in the snapshot in Fig. 3. The grid interval for this model is 2m in x- and z-axes directions. By contrast, the PPW for this computation is approximately 8.57.

The snapshots of the v_x and v_z components at 0.3 s are shown in Fig. 3. The synthetic waveforms of the receivers compared with the GRTM solutions are qualitatively shown in Fig. 4 and

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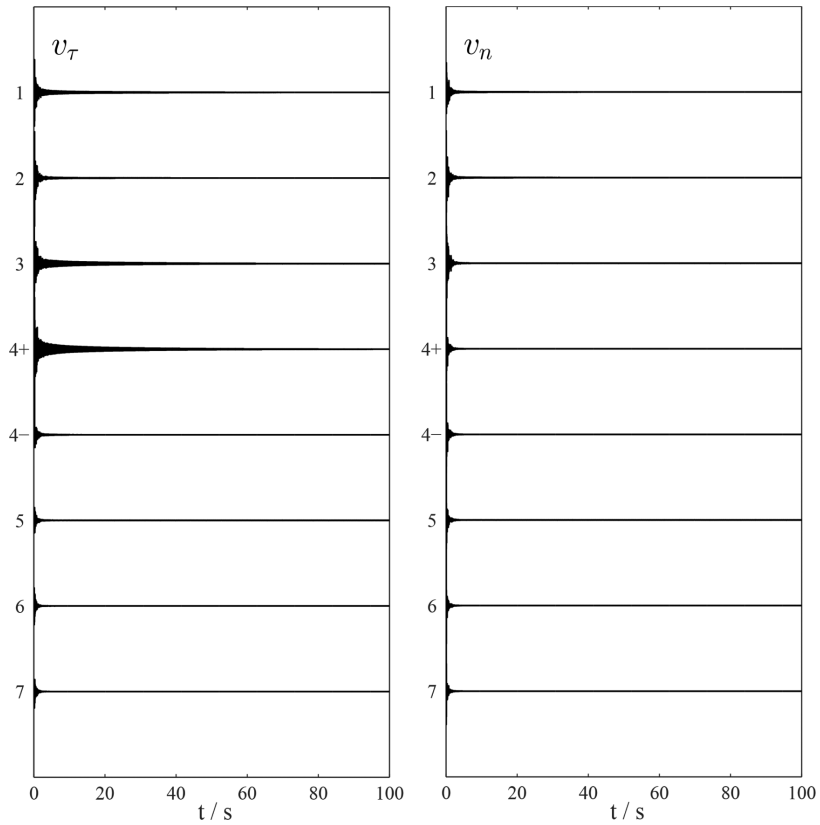


Figure 13. Synthetic v_τ and v_n waveforms for the receivers in Fig. 11 with PPW ≈5. The waveforms last up to 100 s by a time loop of 10^6 times.

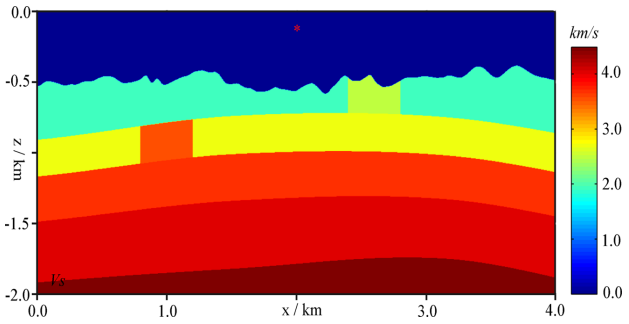


Figure 14. *S*-wave velocity model for the realistic ocean-bottom model. The interface near $z = -0.5$ km is the ocean-bottom topography, which is obtained from the SRTM data. The media parameters are obtained from the Crust 1.0 data. Two heterogeneous media blocks are also added in the model by an increase of 30 per cent of media parameters.

their quantitative comparisons are shown in Table 1. The comparison shows a suitable agreement between the GRTM solutions and curvilinear grid finite-difference solutions. This agreement is due to the maximum value of all the receivers' EMs in Table 1 lower than 7.68 per cent and the maximum value of all the receivers' PMs lower than 1.82 per cent. Moreover, the minimum value of all the receivers' EGs is larger than 9.25, whereas the minimum of all the receivers' PGs is larger than 9.81. These factors are all within the 'excellent' levels of the misfit criteria (Kristeková *et al.* 2009). The relative L_2 errors of the v_x and v_z components are calculated as follows to show the convergence rate of the spatial accuracy:

$$\delta = \frac{||v - v_{\text{reference}}||_{L_2}}{||v_{\text{reference}}||_{L_2}}. \tag{27}$$

The aforementioned equation is plotted in Fig. 5, which supposes that the curvilinear grid finite-difference solutions with PPW = 137.12 are the reference solutions. The said figure shows that the relative errors decrease as the PPW increases, and the algorithm has second-order accuracy. Given the lower-order finite-difference operators near and on the fluid–solid interface, the relative error of the receiver near the interface is larger than that far from the interface (Fig. 5).

Fig. 3 clearly shows that the v_x snapshot is discontinuous across the fluid–solid interface, whereas the v_z snapshot is continuous. This phenomenon is a direct representation of the fluid–solid interface condition (eq. 12), which can also be clearly observed at the receivers on the fluid–solid interface (i.e. receivers 6, 8 and 9) in Fig. 4. The v_x waveforms of receiver 6 (6+ and 6−), receiver 8 (8+ and 8−) and receiver 9 (9+ and 9−) are discontinuous across the interface, whereas their v_z waveforms are continuous.

The conventional FDM in this study implicitly implements the fluid–solid interface condition by effective media parameters (e.g. Graves 1996; Robbert *et al.* 2002; Moczo *et al.* 2007). This FDM is also applied to calculate waveforms in this model. The waveforms computed by the conventional FDM are compared with the GRTM solutions in Fig. 6. Their quantitative comparisons are shown in Table 2. The v_x waveforms (tangential component of the particle velocity) on the fluid–solid interface do not match with the GRTM solutions. The EMs and PMs of the v_x waveforms on the fluid side of the interface (receivers 6+, 8+ and 9+) can be as large as 86.4 per cent. The EMs and PMs do not decrease as the PPW increases (Table 2). This implies that the fluid–solid interface condition (eq. 12) is incorrectly implemented.

The conventional finite-difference solutions of PPW = 32 are compared with the solutions of the curvilinear grid finite difference of PPW = 64 for the receivers on the grid points from receiver 5 to

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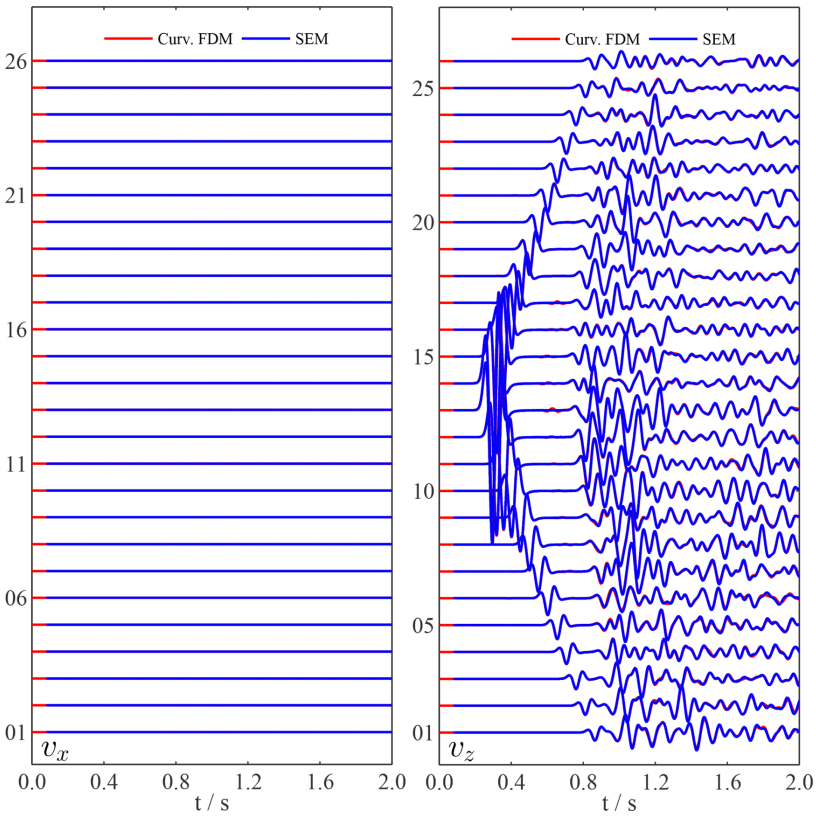


Figure 15. Synthetic v_x and v_z waveforms for the receivers located at sea level from $x = 1.0$ km to $x = 3.0$ km with a spatial interval of 80.0 m in Fig. 14. The red lines are computed by the curvilinear grid FDM (Curv. FDM), whereas the blue lines are computed by SEM.

receiver 7 in this model. This comparison was performed to increase the understanding of the errors near the interface of the conventional FDM. Fig. 7 shows the EM and PM of the solutions on the receivers. The EM and PM values of the v_z component almost remain the same across the fluid–solid interface. For their v_x component, the solutions on the solid side can maintain a lower misfit as low as 2 per cent. By contrast, those on the fluid side can be as much as 84 per cent when the receiver is less than 0.12 wavelength to the fluid–solid interface.

For the flat fluid–solid model, we conclude that the proposed curvilinear FDM can solve the fluid–solid interface problem with second-order accuracy. Because the curvilinear FDM in this paper enforces the numerical solutions to satisfy the exact interface condition, it is also better than the conventional FDM that utilizes effective media parameters (e.g. Graves 1996; Robbert *et al.* 2002; Moczo *et al.* 2007) to implement the interface condition accurately.

4.2 Half-space model with canyon-shaped fluid–solid interface

The second experiment is a half-space with a canyon fluid–solid interface at approximately $z = -200.0$ m in the physical domain. An explosive source is located at (x: 300 m, z: -31.13 m) in the fluid media with $f_c = 10$ Hz and $t_0 = 0.15$ s. The spatial locations of the receivers are shown in Table 3. Receiver positions are selected to coincide with the grid points, which are not evenly spaced in this canyon-shaped fluid–solid interface model. Hence, the receivers are not exactly evenly spaced. The source-receiver geometry and media parameters are shown in the snapshot in Fig. 8. The maximum grid interval for this model is approximately 6.8 m, and the PPW for this computation is approximately 8.82.

The v_x and v_z snapshots at 0.3 s are shown in Fig. 8. Synthetic waveforms of the receivers compared with the SEM solutions are qualitatively shown in Fig. 9 and their quantitative comparisons are listed in Table 3. The comparisons show a suitable agreement between the SEM solutions and curvilinear grid finite-difference solutions. This agreement is achieved because the maximum value of all the receivers' EMs in Table 3 is lower than 6.6 per cent, whereas the maximum value of all receivers' PMs is lower than 2.25 per cent. Moreover, the maximum value of all the receivers' EGs is larger than 9.36, whereas the maximum values of all receivers' PGs is larger than 9.77. These factors are all within the 'excellent' levels of misfit criteria (Kristeková *et al.* 2009).

For this model, we conclude that the Curv. FDM can properly handle the fluid–solid interface and obtain appropriate solutions with a density sampling of 8.82 PPW.

4.3 Sinusoidal fluid–solid interface model

This section demonstrates a half-space model with the sinusoidal shaped fluid–solid interface to test the algorithm in terms of numerical stability. An explosive Ricker source with $f_c = 30.0$ Hz and $t_0 = 0.05$ s is located above and below the interface in two experiments. The source is located at (x: 450.0 m, z: -100.0 m) in the fluid media for the first test, but at (x: 450.0 m, z: -240.0 m) in the solid media for the second test. The receivers are located in a vertical line at $x = 500$ m with a uniform interval of 20 m from receiver 1 ($z = -140$ m) to receiver 7 ($z = -260$ m). The geometry of the sources and receivers in each experiment is shown in Fig. 10.

Fig. 10 also shows the v_x snapshot of the first test and v_z snapshot of the second test. The v_x and v_z snapshots indicate that the topographic fluid–solid interface significantly affects the wavefields. Fig. 11 shows the v_τ and v_n snapshots, which are rotated from v_x and v_z components in Fig. 10 by the formulas in Appendix B. The grid line of the curvilinear coordinates on the interface aligns with the

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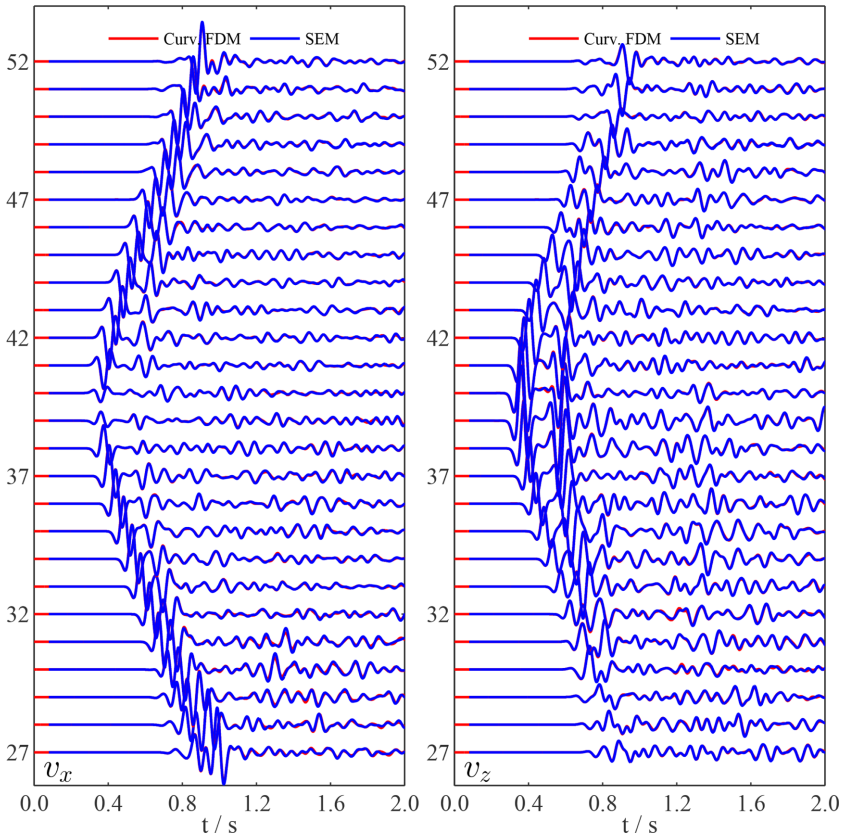


Figure 16. Synthetic v_x and v_z waveforms for the receivers located at $z = -0.4$ km in the ocean from $x = 1.0$ km to $x = 3.0$ km with a spatial interval of 80.0 m in Fig. 14. The red lines are computed by Curv. FDM, whereas the blue lines are computed by SEM.

topographic fluid–solid interface. Thus, the v_τ and v_n components on the interface in Fig. 11 represent the tangential and normal velocity components with respect to the interface. The v_τ component in Fig. 11 may be discontinuous across the fluid–solid interface, whereas the v_n component is continuous.

Fig. 12 shows the synthetic v_τ and v_n waveforms of the receivers in Fig. 11 with different PPWs (i.e. 20, 10 and 5). Their quantitative comparisons are listed in Table 4. The waveforms clearly converge as the PPW increases in Table 4. Receiver 4 is located on the interface, and the v_τ waveforms of the fluid side (receiver 4+) and solid side (receiver 4−) are discontinuous across the interface in Fig. 12. By contrast, the v_n components are continuous. Fig. 13 shows the synthetic v_τ and v_n waveforms of the receivers in Fig. 11. The waveforms last up to 100s by a time loop of 10^6 times. This result shows that the method can be stable for a long time.

We conclude that the solutions of the proposed algorithm for the complex fluid–solid interface not only converge as the PPW increases but also maintain stability for a long time in this complex fluid–solid interface model.

4.4 A realistic ocean-bottom model

This section builds a realistic ocean-bottom model with heterogeneities in the *P*-wave speeds and *S*-wave speeds, as well as densities to validate the proposed curvilinear FDM in a realistic ocean-bottom model by comparing with synthetic waveforms. The realistic 2-D ocean-bottom profile is from (E126.6, N20.5) to (E129.0, N20.5), which are obtained from the shuttle radar topography mission (SRTM) data. The media parameters of the realistic 2-D ocean-bottom model are obtained from Crust 1.0. The 2-D realistic models are compressed into 4.0 km in length and 2.0 km in depth (Fig. 14) to save on computing costs. Two heterogeneous media blocks are also added in the realistic model by an increase of 30 per cent of *P*-wave speed, *S*-wave speed and density to increase the model complexities. Fig. 14 shows the *S*-wave speeds of the model and its two heterogeneous media blocks. An explosive source with $f_c = 6$Hz and $t_0 = 0.2$ s is located at (2.0 km, −0.15 km) in the ocean. The receivers are located in two lines along the *x*-axis on the free surface and in the ocean. The first line contains 26 receivers on the sea level and starts from receiver 1 ($x = 1.0$ km) to receiver 26 ($x = 3.0$ km) with an equal interval of 0.8 km. The second line contains 26 receivers in the fluid ocean at $z = -0.4$ km. This line also starts from receiver 27 ($x = 1.0$ km) to receiver 52 ($x = 3.0$ km) with an equal interval of 0.8 km.

Fig. 15 shows the comparisons of the synthetic waveforms computed by Curv. FDM and SEM for receivers in line 1, whereas Fig. 16 shows those in line 2. The synthetic waveforms are complex in Figs 15 and 16, which are caused by the complex 2-D ocean-bottom topography and heterogeneities in media parameters. The synthetic waveforms computed by Curv. FDM and SEM in Figs 15 and 16 can suitably match. This result hence validates the Curv. FDM in a realistic ocean-bottom model with heterogeneities in the *P*-wave speeds, *S*-wave speeds and densities.

5 CONCLUSIONS

This study extends the Curv. FDM (Zhang & Chen 2006) to simulate seismic wave propagation across a 2-D fluid–solid interface. Seismic waves in fluid media are governed by the acoustic wave equation, whereas those in solid media are governed by the elastic

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wave equation. The fluid–solid interface condition is implemented on the interface. The curvilinear grid is employed to describe the topographic fluid–solid interface because its grid line can align with the interface. As a result, the numerical scattering caused by staircase approximation can be avoided. The wavefield components on the fluid–solid interface may be discontinuous. Hence, the grid points on the fluid–solid interface are split into the fluid and solid points to explicitly implement the fluid–solid interface condition.

This algorithm is validated by comparing with analytical and SEM solutions, and its robustness is also tested by long-time computations. The differences in the synthetic waveforms between this Curv. FDM and the previous FDM are also demonstrated in a flat fluid–solid interface model. This model shows the necessity of this new method for seismic wave propagation in the presence of a fluid–solid interface. The proposed FDM enforces the numerical solutions to satisfy the exact interface condition and it is more accurate than the conventional FDM that uses effective media parameters (e.g. Graves 1996; Robbert *et al.* 2002; Moczo *et al.* 2007) to approximate the interface condition. When the geometry of the fluid–solid interface is complex, the interface needs to be discretized by finer grids to maintain accuracy. However, the discretization would cost additional computing resources. Nevertheless, this issue can be solved by the discontinuous curvilinear grid technique (Li *et al.* 2015) that simultaneously utilizes a finer grid size and a smaller time step for the interface area, and a coarse grid size and a larger time step for areas out of the interface. Future studies can extend this algorithm to 3-D scenario.

ACKNOWLEDGEMENTS

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REFERENCES

Appelö, D. & Petersson, N.A., 2009. A stable finite difference method for the elastic wave equation on complex geometries with free surface, *Commun. Comput. Phys.,* **5**(1), 84–107.

Bayliss, A., Jordan, K.E., LeMesurier, B.J. & Turkel, E., 1986. A fourth-order accurate finite-difference scheme for the computation of elastic waves, *Bull. seism. Soc. Am.,* **76**(4), 1115–1132.

Berenger, J.-P., 1994. A perfectly matched layer for the absorption of electromagnetic waves, *J. Comput. Phys.,* **114,** 185–200.

Bermúdez, A. & Hervella-Nieto, L. R. R., 1999. Finite element computation of three-dimensional elastoacoustic vibrations, *J. Sound Vib.,* **219**(2), 279–306.

Bohlen, T., Kugler, S., Klein, G. & Theilen, F., 2004. 1.5d inversion of lateral variation of Scholte-wave dispersion, *Geophysics,* **69**(2), 330–344.

Bouchon, M., Schultz, C.A. & Toksöz, M.N., 1995. A fast implementation of boundary integral equation methods to calculate the propagation of seismic wave in laterally varying layered media, *Bull. seism. Soc. Am.,* **85**(6), 1679–1687.

Carcione, J.M. & Helle, H.B., 2004. The physics and simulation of wave propagation at the ocean bottom, *Geophysics,* **69**(3), 825–839.

Cerjan, C., Kosloff, D., Kosloff, R. & Reshef, M., 1985. A nonreflecting boundary condition for discrete acoustic and elastic wave equations, *Geophysics,* **50**(4), 705–708.

Chaljub, E., Capdeville, Y. & Vilotte, J.-P., 2003. Solving elastodynamics in a fluid-solid heterogeneous sphere: a parallel spectral element approximation on non-conforming grids, *J. Comput. Phys.,* **187**(2), 457–491.

Chapman, C.H., 2004. *Fundamentals of Seismic Wave Propagation,* Cambridge Univ. Press.

Chen, X.F., 1990. Seismogram synthesis for multi-layered media with irregular interfaces by global generalized reflection/transmission matrices method. I. Theory of two-dimensional SH case, *Bull. seism. Soc. Am.,* **80**(6), 1696–1724.

Chen, X.F., 1995. Seismogram synthesis for multi-layered media with irregular interfaces by global generalized reflection/transmission matrices method. II. Application for 2D SH case, *Bull. seism. Soc. Am.,* **85**(4), 1094–1106.

Chen, X.F., 1996. Seismogram synthesis for multi-layered media with irregular interfaces by global generalized reflection/transmission matrices method. III. Theory of 2D P-SV case, *Bull. seism. Soc. Am.,* **86**(2), 389–405.

Chen, X.F., 1999. Seismograms synthesis in multi-layered half-space media. Part I. Theoretical formulations, *Earthq. Res. China,* **13**(2), 149–174.

Chew, W.C. & Liu, Q.H., 1996. Perfectly matched layers for elastodynamics: a new absorbing boundary condition, *J. Comput. Acoust.,* **4,** 341–359.

Dai, N., Vafidis, A. & Kanasewich, E.R., 1995. Wave propagation in heterogeneous, porous media: a velocity-stress, finite-difference method, *Geophysics,* **60**(2), 327–340.

de Groot-Hedlin, C.D. & Orcutt, J.A., 2001. Excitation of t-phase by seafloor scattering, *J. acoust. Soc. Am.,* **109**(5), 1944–1954.

de Hoop, A.T. & van der Hijden, J.H.M.T., 1983. Generation of acoustic waves by an impulsive line source in a fluid/solid configuration with a plane boundary, *J. acoust. Soc. Am.,* **74**(1), 333–342.

de Hoop, A.T. & van der Hijden, J.H.M.T., 1984. Generation of acoustic waves by an impulsive point source in a fluid/solid configuration with a plane boundary, *J. acoust. Soc. Am.,* **75**(6), 1709–1715.

Ely, G.P., Day, S.M. & Minster, J.-B., 2008. A support-operator method for viscoelastic wave modelling in 3-D heterogeneous media, *Geophys. J. Int.,* **172**(1), 331–344.

Farfour, M. & Yoon, W.J., 2016. A review on multicomponent seismology: a potential seismic application for reservoir characterization, *J. Adv. Res.,* **7,** 515–524.

Feng, K.-A., Teng, C.-H. & Chen, M.-H., 2007. A pseudospectral penalty scheme for 2D isotropic elastic wave computations, *J. Sci. Comput.,* **33**(3), 313–348.

Fichtner, A. & Igel, H., 2008. Efficient numerical surface wave propagation through the optimization of discrete crustal models—a technique based on non-linear dispersion curve matching (DCM), *Geophys. J. Int.,* **173,** 519–533.

Fornberg, B., 1988. The pseudospectral method: accurate representation of interfaces in elastic wave calculations, *Geophysics,* **53**(5), 625–637.

Gao, L., Brossier, R., Pajot, B., Josué, T. & Virieux, J., 2015. An immersed free-surface boundary treatment for seismic wave simulation, *Geophysics,* **80**(5), 193–209.

Ge, Z. & Chen, X., 2007. Wave propagation in irregularly layered elastic models: a boundary element approach with a global reflection/transmission matrix propagator, *Bull. seism. Soc. Am.,* **97**(3), 1025–1031.

Graves, R.W., 1996. Simulating seismic wave propagation in 3D elastic media using staggered-grid finite differences, *Bull. seism. Soc. Am.,* **86**(4), 1091–1106.

Greaves, R.J. & Stephen, R.A., 2000. Low-grazing-angle momostatic acoustic reverberation from rough and heterogeneous seafloors, *J. acoust. Soc. Am.,* **108**(3), 1013–1025.

Hestholm, S. & Ruud, B., 1994. 2-D finite-difference elastic wave modeling including surface topography, *Geophys. Prospect.,* **42**(5), 371–390.

Hestholm, S. & Ruud, B., 1998. 3-D finite-difference elastic wave modeling including surface topography, *Geophysics,* **63**(2), 613–622.

Hixon, R., 1997. On increasing the accuracy of MacCormack schemes for aeroacoustic applications, in *Proceedings of the 3rd AIAA/CESA Aeroacoustics Conference, AIAA Paper 97-1586,* pp. 29–39, Atlanta, GA.

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Hung, S.-H. & Forsyth, D.W., 1998. Modelling anisotropic wave propagation in oceanic inhomogeneous structure using the parallel multidomain pseudo-spectral method, *Geophys. J. Int.,* **133**(3), 726–740.

Igel, H., Mora, P. & Riollet, B., 1995. Anisotropic wave propagation through finite-difference grids, *Geophysics,* **60**(4), 1203–1216.

Jamet, G., Guennou, C., Guillon, L., Mazoyer, C. & Royer, J.-Y., 2013. T-wave generation and propagation: a comparison between data and spectral element modeling, *J. acoust. Soc. Am.,* **134**(4), 3376–3385.

Käser, M. & Dumbser, M., 2008. A highly accurate discontinuous Galerkin method for complex interfaces between solid and moving fluids, *Geophysics,* **73**(3), T23–T35.

Käser, M., Hermann, V. & Puente, J.D.L., 2008. Quantitative accuracy analysis of the discontinuous Galerkin method for seismic wave propagation, *Geophys. J. Int.,* **173,** 990–999.

Komatitsch, D., Barnes, C. & Tromp, J., 2000. Wave propagation near a fluid-solid interface: a spectral-element approach, *Geophysics,* **65**(2), 623–631.

Kreiss, H.-O. & Petersson, N.A., 2006. An embedded boundary method for the wave equation with discontinuous coefficients, *SIAM J. Sci. Comput.,* **28**(6), 2054–2074.

Kristek, J., Moczo, P. & Galis, M., 2010. Stable discontinuous staggered grid in the finite-difference modelling of seismic motion, *Geophys. J. Int.,* **183**(3), 1401–1407.

Kristeková, M., Kristek, J., Moczo, P. & Day, S.M., 2006. Misfit criteria for quantitative comparison of seismograms, *Bull. seism. Soc. Am.,* **96**(5), 1836–1850.

Kristeková, M., Kristek, J. & Moczo, P., 2009. Time-frequency misfit and goodness-of-fit criteria for quantitative comparison of time signals, *Geophys. J. Int.,* **178**(2), 813–825.

Kugler, S., Bohlen, T., Forbriger, T., Bussat, S. & Klein, G., 2007. Scholte-wave tomography for shallow-water marine sediments, *Geophys. J. Int.,* **168**(2), 551–570.

Landau, L.D. & Lifshitz, E.M., 1959. *Fluid Mechanics,* 2nd edn, Pergamon Press.

Li, H., Zhang, W., Zhang, Z. & Chen, X., 2015. Elastic wave finite-difference simulation using discontinuous curvilinear grid with non-uniform time step: two-dimensional case, *Geophys. J. Int.,* **202,** 102–118.

Lombard, B. & Piraux, J., 2004. Numerical treatment of two-dimensional interfaces for acoustic and elastic waves, *J. Comput. Phys.,* **195**(1), 90–116.

Lombard, B., Piraux, J., Gélis, C. & Virieux, J., 2008. Free and smooth boundaries in 2-D finite-difference schemes for transient elastic waves, *Geophys. J. Int.,* **172**(1), 252–261.

Luco, J.E. & Apsel, J.R., 1983. On the Green's function for a layered half-space. Part I, *Bull. seism. Soc. Am.,* **73**(4), 909–927.

Magnier, S.-A., Mora, P. & Tarantola, A., 1994. Finite-difference on minimal grids, *Geophysics,* **59**(9), 1435–1443.

Moczo, P., Kristek, J., Vavryčuk, V., Archuleta, R.J. & Halada, L., 2002. 3D heterogeneous staggered-grid finite-difference modeling of seismic motion with volume harmonic and arithmetic averaging of elastic moduli and densities, *Bull. seism. Soc. Am.,* **92**(8), 3042–3066.

Moczo, P., Robertsson, J.O.A. & Eisner, L., 2007. The finite-difference time-domain method for modeling of seismic wave propagation, *Adv. Geophys.,* **48,** 421–516.

Muir, F., Dellinger, J., Etgen, J. & Nichols, D., 1992. Modelling elastic fields across irregular boundaries, *Geophysics,* **57**(9), 1189–1193.

Muyzert, E., 2007a. Seabed property estimation from ambient-noise recording: compliance and Scholte wave phase-velocity measurements, *Geophysics,* **72**(2), 21–26.

Muyzert, E., 2007b. Seabed property estimation from ambient-noise recording: Part 2—Scholte wave spectral-ratio inversion, *Geophysics,* **72**(4), 47–53.

Noguchi, S., Maeda, T. & Furumura, T., 2007. FDM simulation of an anomalous later phase from the Japan trench subduction zone earthquake, *Pure appl. Geophys.,* **170,** 95–108.

Okal, E.A., 2008. The generation of T waves by earthquakes, *Adv. Geophys.,* **49,** 1–65.

Okamoto, T. & Takenaka, H., 2005. Fluid-solid boundary implementation in the velocity-stress finite difference method, *Zisin,* **57**(2), 355–364.

Padilla, F., Billy, M.D. & Quentin, G., 1999. Theoretical and experimental studies of surface waves on solid-fluid interfaces when the value of the fluid sound velocity in located between the shear and the longitudinal ones in the solid, *J. acoust. Soc. Am.,* **106**(2), 666–673.

Pérez-Ruiz, J.A., Luzó, F. & García-Jerez, A., 2007. Scattering of elastic waves in cracked media using a finite difference method, *Stud. Geophys. Geod.,* **51,** 59–88.

Puente, J.D.L., Ferrer, M., Hanzich, M., Castillo, J.E. & Cela, J.M., 2014. Mimetic seismic wave modeling including topography on deformed staggered grids, *Geophysics,* **79**(3), 125–141.

Qian, Z.-H. & Yamanaka, H., 2012. An efficient approach for simulating seismoacoustic scattering due to an irregular fluid-solid interface in multilayered media, *Geophys. J. Int.,* **189**(1), 524–540.

Robbert, V.V., Robertsson, J.O.A. & Chapman, C.H., 2002. Finite-difference modeling of wave propagation in a fluid-solid configuration, *Geophysics,* **67**(2), 618–624.

Robertsson, J.O.A., 1996. A numerical free-surface condition for elastic/viscoelastic finite-difference modeling in the presence of topography, *Geophysics,* **61**(6), 1921–1934.

Robertsson, J.O.A. & Levander, A., 1995. A numerical study of seafloor scattering, *J. acoust. Soc. Am.,* **97**(3), 3532–3546.

Sun, Y.-C., Zhang, W. & Xiaofei, C., 2016. Seismic-wave modelling in the presence of surface topography in 2D general anisotropic media by a curvilinear grid finite-difference method, *Bull. seism. Soc. Am.,* **106**(3), 1036–1054.

Tarrass, I., Giraud, L. & Thore, P., 2011. New curvilinear scheme for elastic wave propagation in presence of curved topography, *Geophys. Prospect.,* **59**(5), 889–906.

Tessmer, E., Kosloff, D. & Alfred, B., 1992. Elastic wave propagation simulation in the presence of surface topography, *Geophys. J. Int.,* **108**(2), 621–632.

Thompson, J.F., Warsi, Z.U.A. & Mastin, C.W., 1985. *Numerical Grid Generation – Foundations and Applications,* North Holland Publishing Co.

Vdfidus, A., Abramovici, F. & Kanasewich, E.R., 1992. Elastic wave propagation using fully vectorized high order finite-difference algorithms, *Geophysics,* **57**(2), 218–232.

Voinovich, P., Merlen, A. & Timofeev, E. K. T., 2003. A Godunov-type finite-volume scheme for unified solid-liquid elastodynamics on arbitrary two-dimensional grids, *Shock Waves,* **13**(3), 221–230.

Wang, L. & Liang, C., 2006. A new implementation of CFS-PML for ADI-FDTD method, *Microw. Opt. Technol. Lett.,* **48**(10), 1924–1928.

Wilcox, L.C., Stadler, G., Burstedde, C. & Ghattas, O., 2010. A high-order discontinuous Galerkin method for wave propagation through coupled elastic-acoustic media, *J. Comput. Phys.,* **229,** 9373–9396.

Xie, X.B. & Yao, Z.X., 1988. P-SV wave responses for a point source in two-dimensional heterogeneous media: finite-difference approximation, *Chin. J. Geophys.,* **31**(5), 473–493.

Zhang, J., 2004. Wave propagation across fluid-solid interfaces: a grid method approach, *Geophys. J. Int.,* **159**(1), 240–252.

Zhang, W. & Chen, X.F., 2006. Traction image method for irregular free surface boundaries in finite difference seismic wave simulation, *Geophys. J. Int.,* **167**(1), 337–353.

Zhang, W. & Yang, S., 2010. Unsplit complex frequency-shift PML implementation using auxiliary differential equation for seismic wave modelling, *Geophysics,* **75**(4), 141–154.

Zhang, W., Shen, Y. & Zhao, L., 2012a. Three-dimensional anisotropic seismic wave modelling in spherical coordinates by a collocated-grid finite-difference method, *Geophys. J. Int.,* **188**(3), 1359–1381.

Zhang, W., Zhang, Z.G. & Chen, X.F., 2012b. Three-dimensional elastic wave numerical modelling in the presence of surface topography by a collocated-grid finite-difference method on curvilinear grids, *Geophys. J. Int.,* **190**(1), 358–378.

Zhang, Z.G., Zhang, W., Li, H. & Chen, X., 2013. Stable discontinuous grid implementation for collocated-grid finite-difference seismic wave modelling, *Geophys. J. Int.,* **192**(3), 1179–1188.

Zhang, Z.G., Zhang, W. & Chen, X., 2014a. Complex frequency-shifted multi-axial perfectly matched layer for elastic wave modelling on curvilinear grids, *Geophys. J. Int.,* **198**(1), 140–153.

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Zhang, Z.G., Zhang, W. & Chen, X.F., 2014b. Three-dimensional curved grid finite-difference modelling for non-planar rupture dynamics, *Geophys. J. Int.,* **199**(2), 860–879.

Zheng, Y., Fang, X., Liu, J. & Fehler, M.C., 2013. Scholte waves generated by seafloor topography, arXiv:1306.4383.

Zhu, J. & Popovics, J.S., 2004. Leaky Rayleigh and Scholte waves at the fluid-solid interface subjected to transient point loading, *J. acoust. Soc. Am.,* **116**(4), 2101–2110.

APPENDIX A: COEFFICIENTS MATRICES

For acoustic wave equation the coefficients matrices $\boldsymbol{W}$, $\boldsymbol{A}$, $\boldsymbol{B}$ and $\boldsymbol{F}$ of eq. (7) can be written as

$$\boldsymbol{W} = (v_x^f \quad v_z^f \quad p)^T \tag{A1}$$

$$\boldsymbol{A} = \begin{pmatrix} 0 & 0 & -\partial_x\xi/\rho_f \\ 0 & 0 & -\partial_z\xi/\rho_f \\ -\kappa\partial_x\xi & -\kappa\partial_z\xi & 0 \end{pmatrix} \tag{A2}$$

$$\boldsymbol{B} = \begin{pmatrix} 0 & 0 & -\partial_x\zeta/\rho_f \\ 0 & 0 & -\partial_z\zeta/\rho_f \\ -\kappa\partial_x\zeta & -\kappa\partial_z\zeta & 0 \end{pmatrix} \tag{A3}$$

and for elastic wave equation the coefficients matrices $\boldsymbol{W}$, $\boldsymbol{A}$, $\boldsymbol{B}$ and $\boldsymbol{F}$ of eq. (7) can be written as

$$\boldsymbol{W} = (v_x^s \quad v_z^s \quad \sigma_{xx} \quad \sigma_{zz} \quad \sigma_{xz})^T \tag{A4}$$

$$\boldsymbol{A} = \begin{pmatrix} 0 & 0 & \partial_x\xi/\rho_s & 0 & \partial_z\xi/\rho_s \\ 0 & 0 & 0 & \partial_z\xi/\rho_s & \partial_x\xi/\rho_s \\ \chi\partial_x\xi & \lambda\partial_z\xi & 0 & 0 & 0 \\ \lambda\partial_x\xi & \chi\partial_z\xi & 0 & 0 & 0 \\ \mu\partial_z\xi & \mu\partial_x\xi & 0 & 0 & 0 \end{pmatrix} \tag{A5}$$

$$\boldsymbol{B} = \begin{pmatrix} 0 & 0 & \partial_x\zeta/\rho_s & 0 & \partial_z\zeta/\rho_s \\ 0 & 0 & 0 & \partial_z\zeta/\rho_s & \partial_x\zeta/\rho_s \\ \chi\partial_x\zeta & \lambda\partial_z\zeta & 0 & 0 & 0 \\ \lambda\partial_x\zeta & \chi\partial_z\zeta & 0 & 0 & 0 \\ \mu\partial_z\zeta & \mu\partial_x\zeta & 0 & 0 & 0 \end{pmatrix}. \tag{A6}$$

APPENDIX B: ROTATION TRANSFORMATION

Because the ξ-axis is assumed parallel to the fluid–solid interface, the unit vectors perpendicular and parallel to the fluid–solid interface can be stated as

$$\boldsymbol{n} = \frac{\partial_x\zeta}{s}\boldsymbol{i} + \frac{\partial_z\zeta}{s}\boldsymbol{k} \tag{B1a}$$

$$\boldsymbol{\tau} = \frac{\partial_z\zeta}{s}\boldsymbol{i} - \frac{\partial_x\zeta}{s}\boldsymbol{k} \tag{B1b}$$

where $\boldsymbol{i}$ and $\boldsymbol{k}$ are the unit vectors of x-axis and z-axis of Cartesian coordinates and $s = \sqrt{(\partial_x\zeta)^2 + (\partial_z\zeta)^2}$. A rotation matrix $\boldsymbol{R}$ is defined as

$$\boldsymbol{R} = \begin{pmatrix} \partial_z\zeta/s & -\partial_x\zeta/s \\ \partial_x\zeta/s & \partial_z\zeta/s \end{pmatrix} \tag{B2}$$

where $\boldsymbol{R}$ is an anti-symmetric matrix and $\boldsymbol{R}\boldsymbol{R}^T = \boldsymbol{I}$.

Take $\boldsymbol{v} = (v_x, v_z)$ as the particle velocity in Cartesian coordinate and $\boldsymbol{v}' = (v_\tau, v_n)$ the particle velocity in local coordinate. The relation between $\boldsymbol{v}$ and $\boldsymbol{v}'$ are

$$\boldsymbol{v}' = \boldsymbol{R}\boldsymbol{v} \tag{B3a}$$

$$\boldsymbol{v} = \boldsymbol{R}^T\boldsymbol{v}' \tag{B3b}$$

written in full length

$$v_\tau = \frac{\partial_z\zeta}{s}v_x - \frac{\partial_x\zeta}{s}v_z \tag{B4a}$$

$$v_n = \frac{\partial_x\zeta}{s}v_x + \frac{\partial_z\zeta}{s}v_z \tag{B4b}$$

and

$$v_x = \frac{\partial_z\zeta}{s}v_\tau + \frac{\partial_x\zeta}{s}v_n \tag{B5a}$$

$$v_z = -\frac{\partial_x\zeta}{s}v_\tau + \frac{\partial_z\zeta}{s}v_n. \tag{B5b}$$

Take $\boldsymbol{\sigma}$ as the stress tensor in Cartesian coordinate and $\boldsymbol{\sigma}'$ in local coordinate. The relation between $\boldsymbol{\sigma}$ and $\boldsymbol{\sigma}'$ is

$$\boldsymbol{\sigma}' = \boldsymbol{R}\boldsymbol{\sigma}\boldsymbol{R}^T \tag{B6a}$$

$$\boldsymbol{\sigma} = \boldsymbol{R}^T\boldsymbol{\sigma}'\boldsymbol{R} \tag{B6b}$$

where

$$\boldsymbol{\sigma}' = \begin{pmatrix} \sigma_{\tau\tau} & \sigma_{n\tau} \\ \sigma_{\tau n} & \sigma_{nn} \end{pmatrix} \tag{B7}$$

and

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{xx} & \sigma_{xz} \\ \sigma_{xz} & \sigma_{zz} \end{pmatrix} \tag{B8}$$

written in full length

$$\sigma_{\tau\tau} = \frac{\partial_z\zeta\partial_z\zeta}{s^2}\sigma_{xx} - 2\frac{\partial_x\zeta\partial_z\zeta}{s^2}\sigma_{xz} + \frac{\partial_x\zeta\partial_x\zeta}{s^2}\sigma_{zz} \tag{B9a}$$

$$\sigma_{nn} = \frac{\partial_x\zeta\partial_x\zeta}{s^2}\sigma_{xx} + 2\frac{\partial_x\zeta\partial_z\zeta}{s^2}\sigma_{xz} + \frac{\partial_z\zeta\partial_z\zeta}{s^2}\sigma_{zz} \tag{B9b}$$

$$\sigma_{\tau n} = \sigma_{n\tau} = \frac{\partial_x\zeta\partial_z\zeta}{s^2}\sigma_{xx} + \frac{\partial_z\zeta\partial_z\zeta - \partial_x\zeta\partial_x\zeta}{s^2}\sigma_{xz} - \frac{\partial_x\zeta\partial_z\zeta}{s^2}\sigma_{zz} \tag{B9c}$$

and

$$\sigma_{xx} = \frac{\partial_z\zeta\partial_z\zeta}{s^2}\sigma_{\tau\tau} + 2\frac{\partial_x\zeta\partial_z\zeta}{s^2}\sigma_{n\tau} + \frac{\partial_x\zeta\partial_x\zeta}{s^2}\sigma_{nn} \tag{B10a}$$

$$\sigma_{zz} = \frac{\partial_x\zeta\partial_x\zeta}{s^2}\sigma_{\tau\tau} - 2\frac{\partial_x\zeta\partial_z\zeta}{s^2}\sigma_{n\tau} + \frac{\partial_z\zeta\partial_z\zeta}{s^2}\sigma_{nn} \tag{B10b}$$

$$\sigma_{xz} = -\frac{\partial_x\zeta\partial_z\zeta}{s^2}\sigma_{\tau\tau} + \frac{\partial_z\zeta\partial_z\zeta - \partial_x\zeta\partial_x\zeta}{s^2}\sigma_{n\tau}. \tag{B10c}$$

For the case of acoustic wave equation ($\sigma_{xx} = \sigma_{zz} = -p$ and $\sigma_{xz} = 0$), $\boldsymbol{\sigma}$ and $\boldsymbol{\sigma}'$ can be expressed as

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{xx} & \sigma_{xz} \\ \sigma_{xz} & \sigma_{zz} \end{pmatrix} = \begin{pmatrix} -p & 0 \\ 0 & -p \end{pmatrix} \tag{B11}$$

and

$$\boldsymbol{\sigma}' = \begin{pmatrix} \sigma_{\tau\tau} & \sigma_{\tau n} \\ \sigma_{n\tau} & \sigma_{nn} \end{pmatrix} = \begin{pmatrix} -p & 0 \\ 0 & -p \end{pmatrix}. \tag{B12}$$

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